



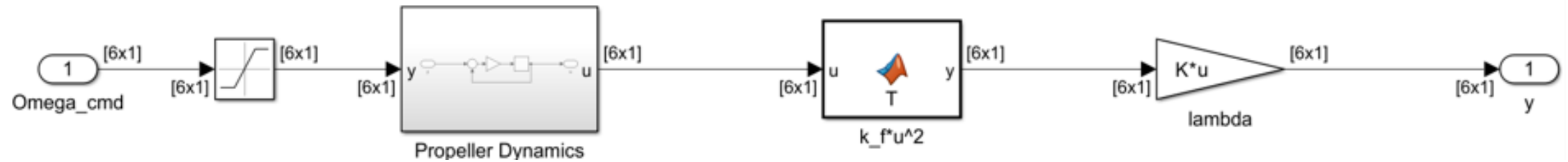
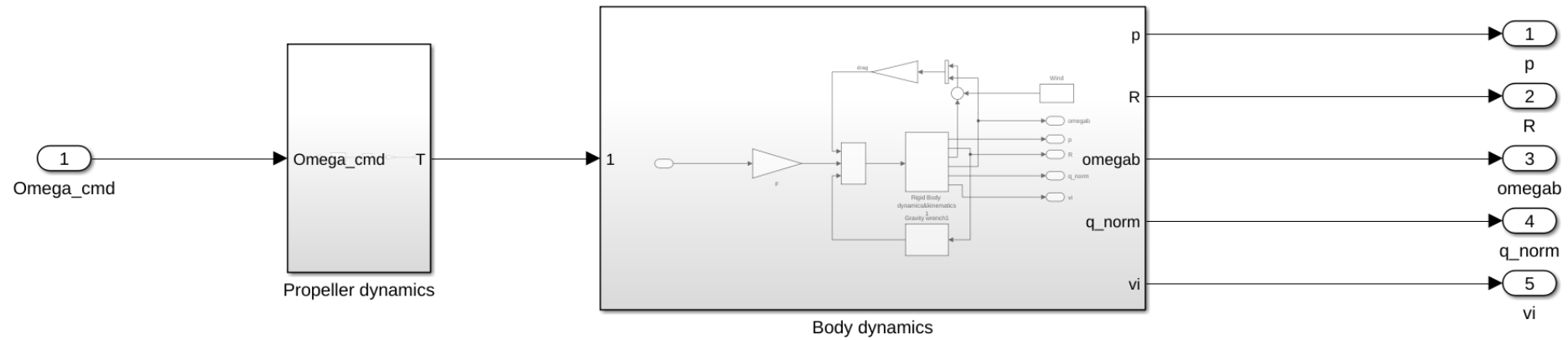
POLITECNICO
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Project Presentation

Adaptive and Autonomous Aerospace Systems
A.Y. 2025/2026

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Task 1: Simulation Model Implementation



Task 1: Trim analytically at $p=[0 \ 0 \ 1]'$

$$T_r = \lambda_r k_f \Omega_r^2 \xrightarrow{\lambda_r = 1 \ \forall r} \Omega_r = \sqrt{\frac{T_r}{k_f}}$$

UAV Dynamics: $M\dot{v}_b + v_b^x M v_b = -D v_b + w_g + Fu$

$$\text{Trim: } v_b = 0 \ \forall t \Rightarrow \dot{v}_b = 0$$

$$-w_g = Fu$$



Task 1: Trim analytically at $p=[0 \ 0 \ 1]'$

$$\begin{bmatrix} \text{mgR}^T \mathbf{e}_3 \\ \text{gSR}^T \mathbf{e}_3 \end{bmatrix} = \begin{bmatrix} 0 & \cdots \\ 0 & \cdots \\ 1 & \cdots \\ b \sin \gamma_1 & \cdots \\ -b \cos \gamma_1 & \cdots \\ -\sigma_1 \xi_1 & \cdots \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \\ T_3 \\ T_4 \\ T_5 \\ T_6 \end{bmatrix}$$

$$\gamma_i = \frac{2\pi}{N_r} (i - 1) \quad \forall i \in (1, \dots, N_r), \quad N_r = 6$$

$$\mathbf{S} = \mathbf{0}$$

$$\mathbf{R} = \mathbf{I}_3$$

$$\xi_i = (-1)^{i-1}$$



Task 1: Trim analytically at $p=[0 \ 0 \ 1]'$

4 equations, 6 unknowns

$$T_2 + T_3 - T_5 - T_6 = 0$$

$$T_1 + \frac{1}{2}T_2 - \frac{1}{2}T_3 - T_4 - \frac{1}{2}T_5 + \frac{1}{2}T_6 = 0$$

$$T_1 - T_2 + T_3 - T_4 + T_5 - T_6 = 0$$

$$mg = \sum_i T_i$$



$$T_1 = T_4$$

$$T_2 = T_5$$

$$T_6 = T_3$$

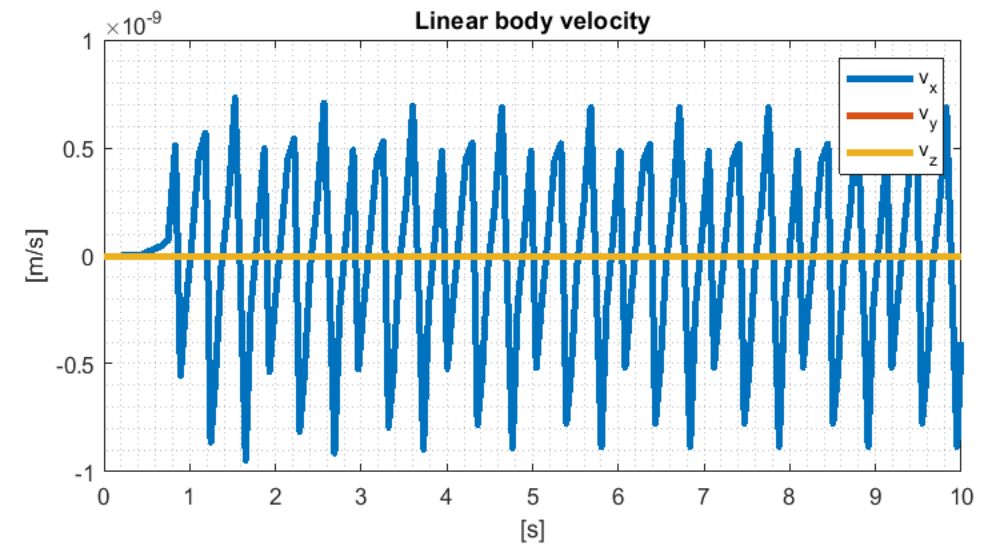
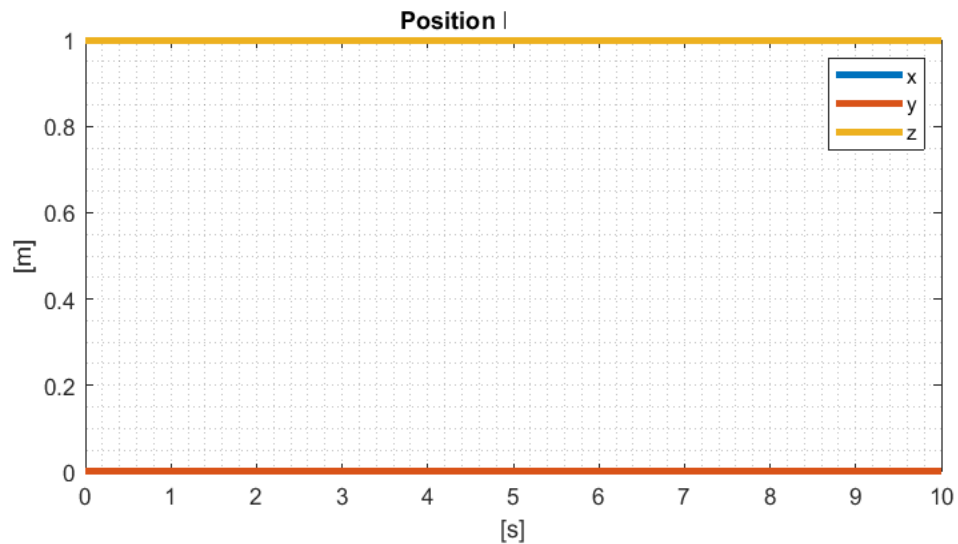
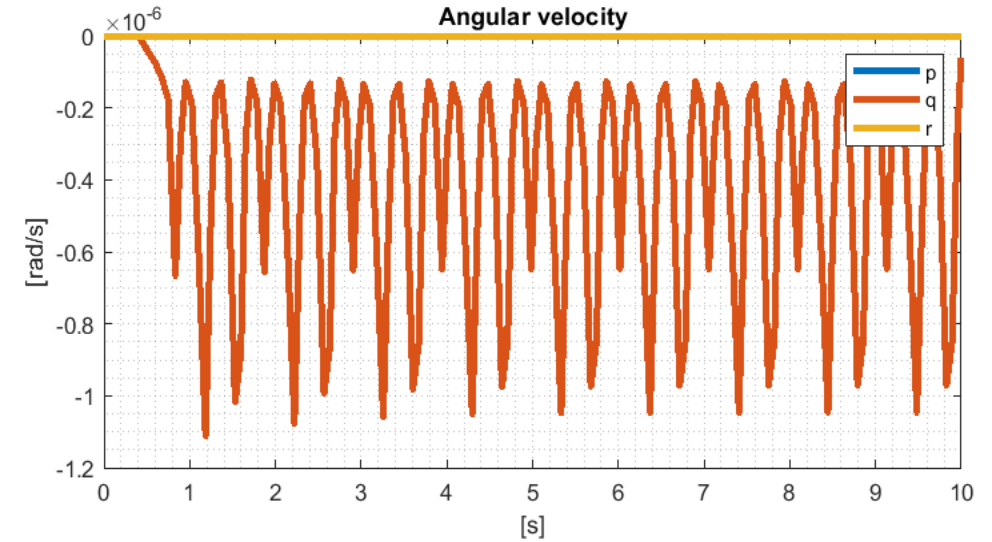
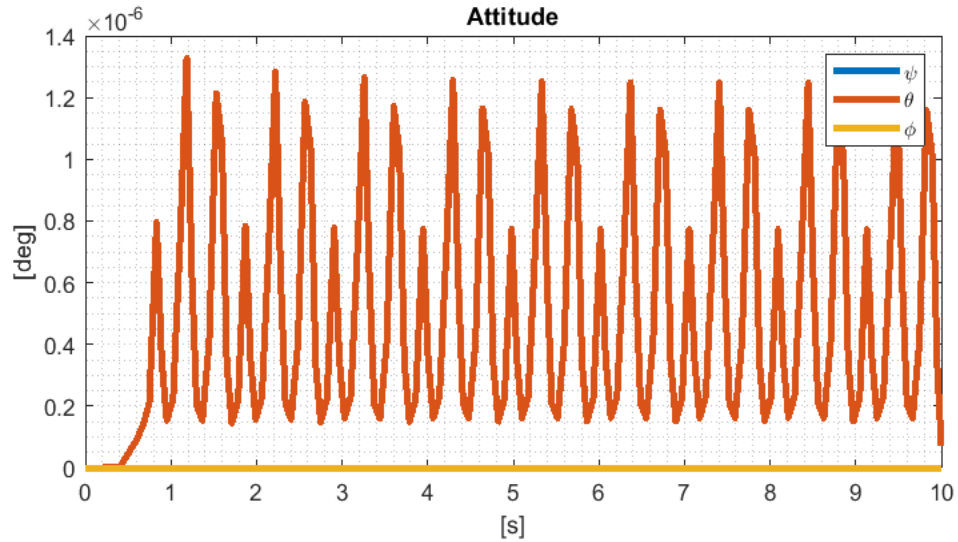
$$T_i = T$$



$$\Omega_{r,cmd} = \sqrt{\frac{mg}{6k_f}}$$



Task 1: Trim analytically at $p=[0 \ 0 \ 1]'$



Task 1: Trim at $v_i=[2\ 0\ 0]'$ m/s

UAV Dynamics: $M\dot{v}_b + v_b^x M v_b = -D v_b + w_g + F u$

$$\text{Trim: } v_b = \begin{bmatrix} 2 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad \forall t, \dot{v}_b = 0$$

$$-D v_b + w_g + F u = 0$$



Task 1: Trim at $v_i=[2\ 0\ 0]'$ m/s

$$\begin{bmatrix} -D_f \bar{v} \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} -mg \sin \theta \\ mg \sin \phi \cos \theta \\ mg \cos \phi \cos \theta \\ 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \sum_i T_i \\ b \sum_i \sin \gamma_i T_i \\ -b \sum_i \cos \gamma_i T_i \\ \sum_i \sigma_i \xi_i T_i \end{bmatrix} = 0$$

Same equations as before

+

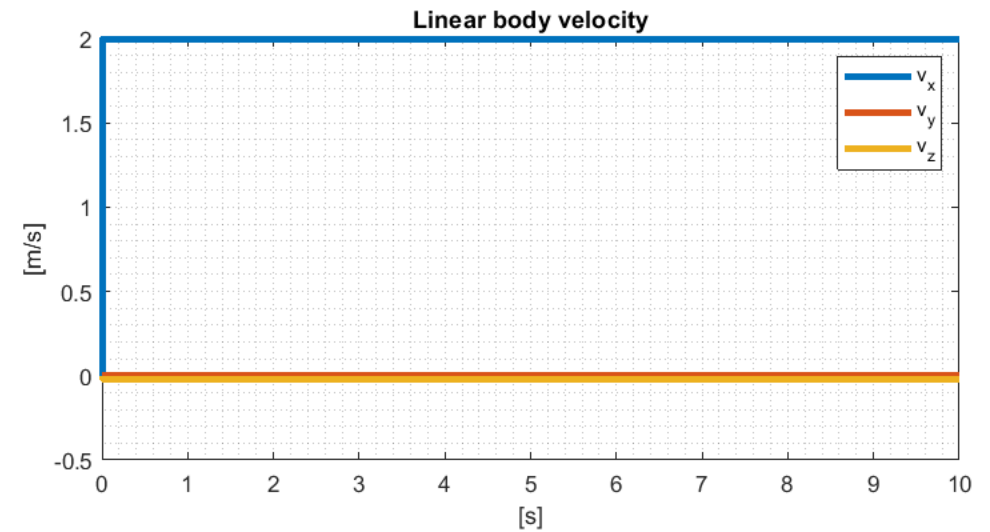
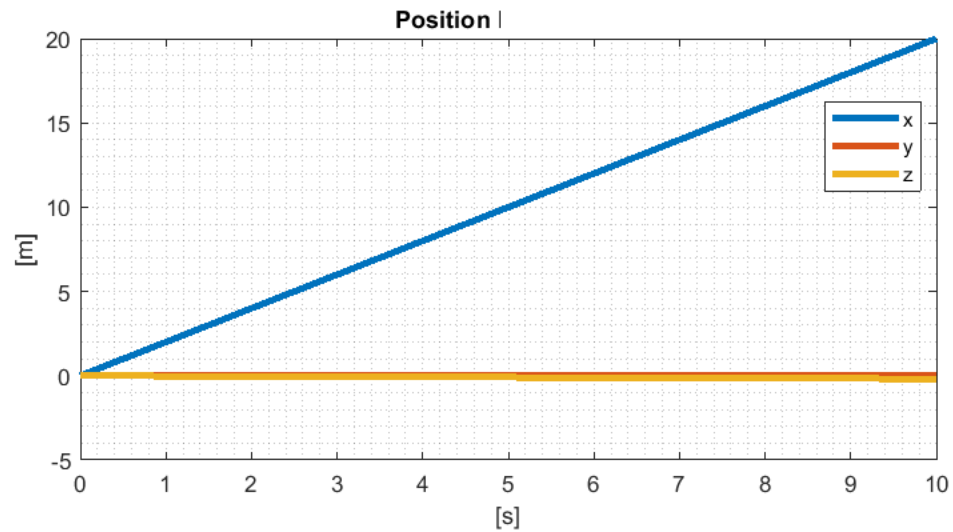
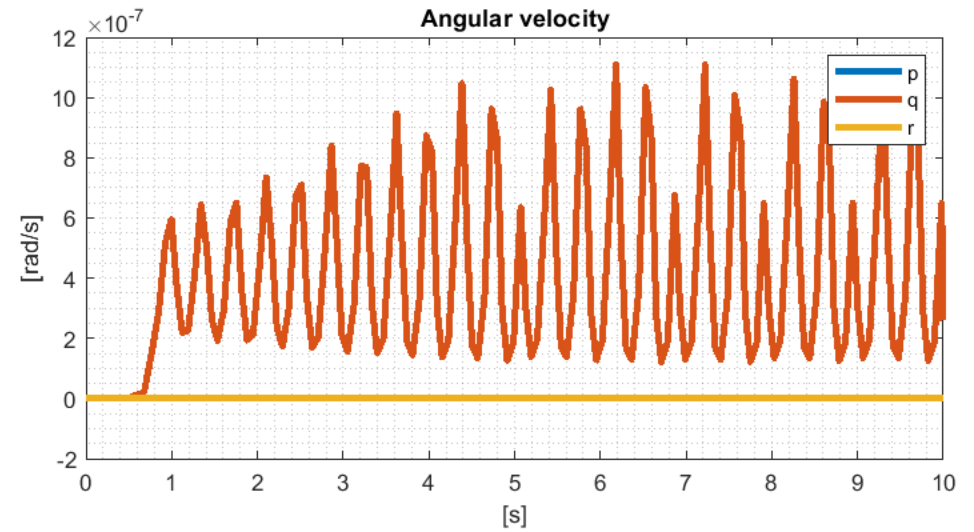
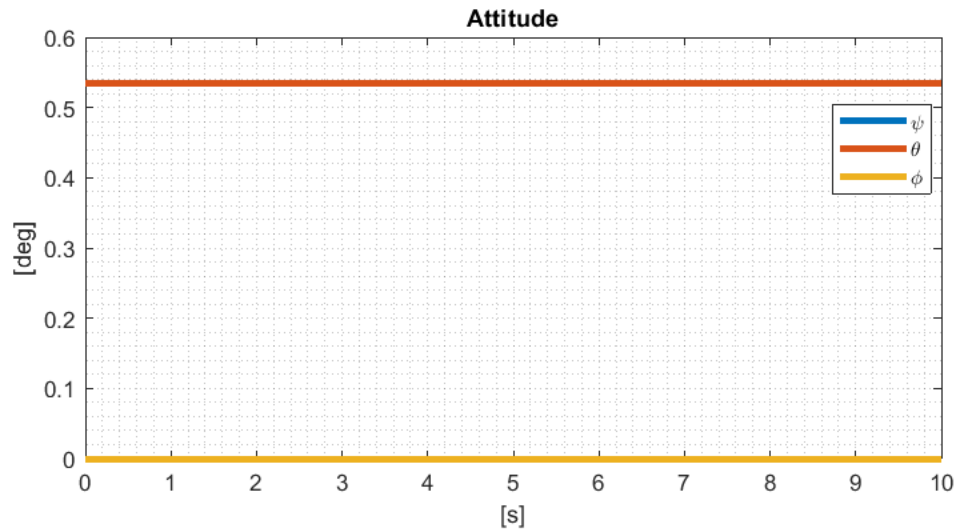
$$\theta = \arcsin\left(\frac{D_f \bar{v}}{mg}\right)$$

$$\phi = 0$$

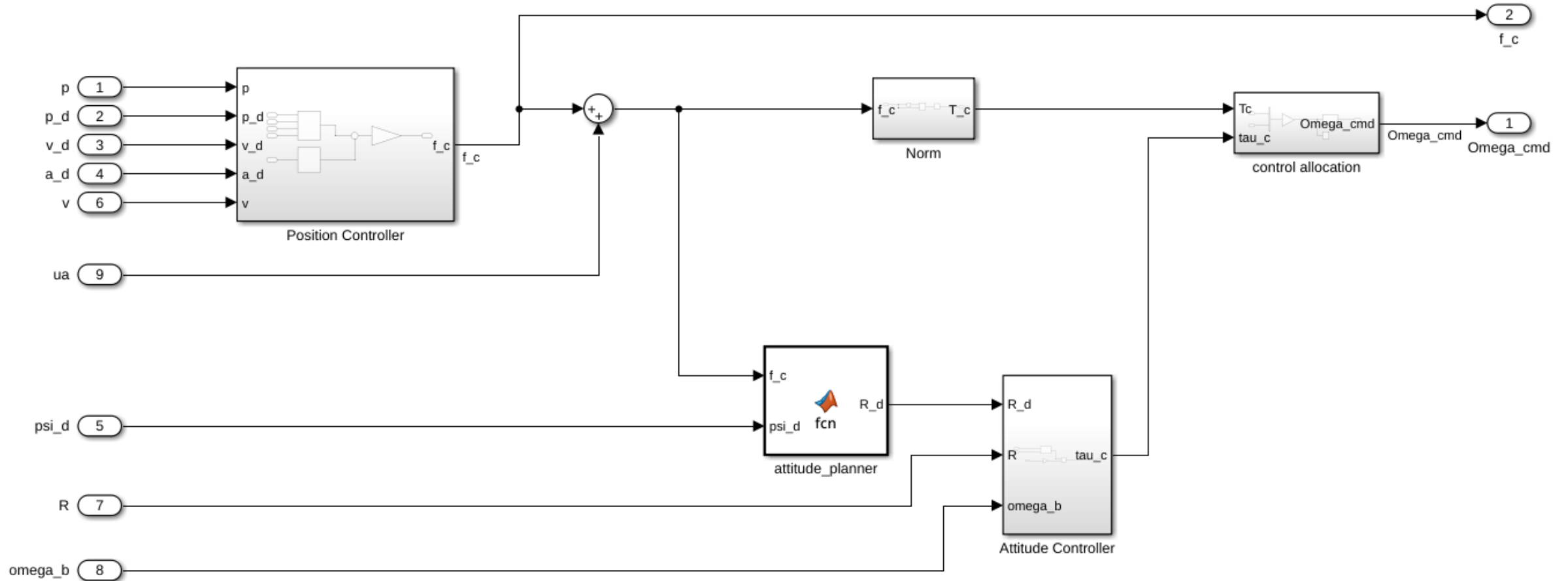
$$\Omega_{r,cmd} = \sqrt{\frac{mg \cos \theta}{6k_f}}$$



Task 1: Trim at $v_i=[2\ 0\ 0]'$ m/s



Task 2: Baseline controller implementation



Task 2: Position controller

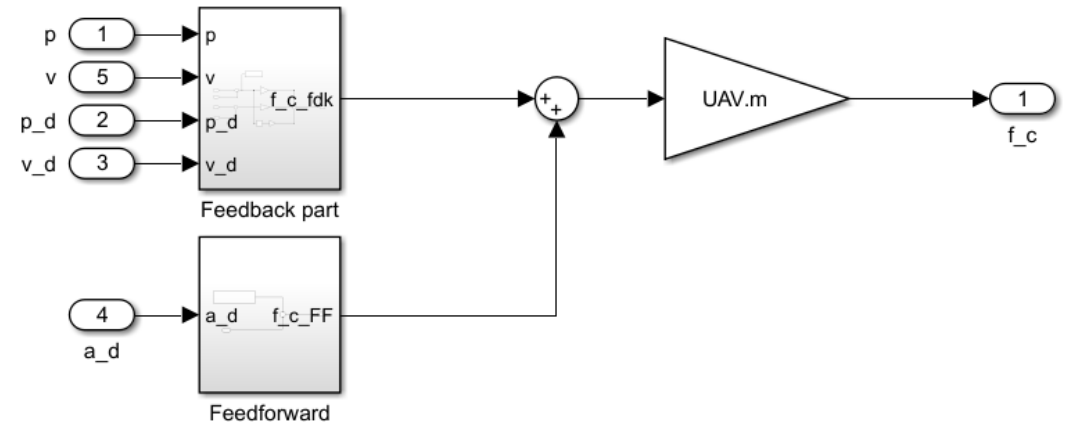
Decoupled position dynamics

$$\begin{aligned}\dot{p} &= v_i \\ m\dot{v}_i &= -mge_3 + f_e + f_c\end{aligned}$$

Control law

$$f_c := \underbrace{-m(K_p e_p + K_v e_v + K_i e_i)}_{\text{feedback}} + \underbrace{m(g e_3 + \dot{v}_d)}_{\text{feedforward}}$$

GPS for position measurement are modeled as a delay of 30 ms



Task 2: Attitude controller

Decoupled attitude dynamics

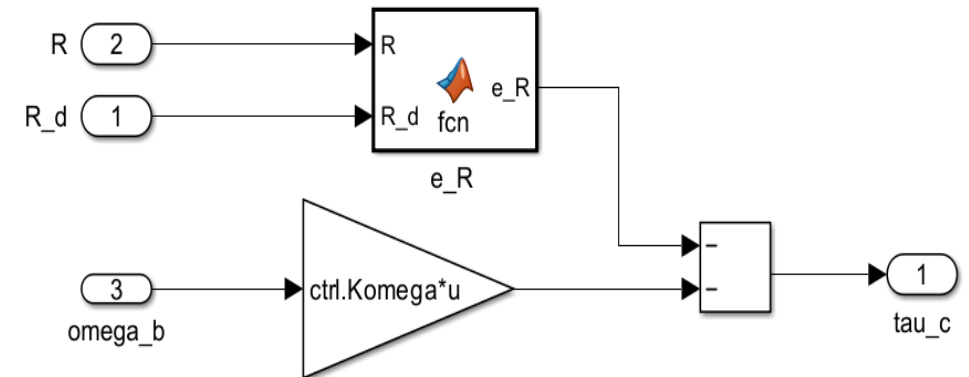
$$\dot{R} = R \omega_b^\times$$

Geometric control law

$$\tau_c \approx -K_\omega \omega_b - e_R$$

$$e_R = K_{R1}(R_e^T e_1 \times e_1) + K_{R2}(R_e^T e_2 \times e_2) + K_{R3}(R_e^T e_3 \times e_3)$$
$$R_e = R_d^T R$$

Gyroscope for attitude measurement are modeled as a delay of 10 ms



Task 2: Attitude planner

$$h_d = [\cos(\psi_d) \sin(\psi_d) 0]^T$$

$$k_p = \frac{f_c}{|f_c|}$$

$$j_p = \frac{k_p \times h_d}{|k_p \times h_d|}$$

$$i_p = j_p \times k_p$$

$$R = [i_p \ j_p \ k_p]$$

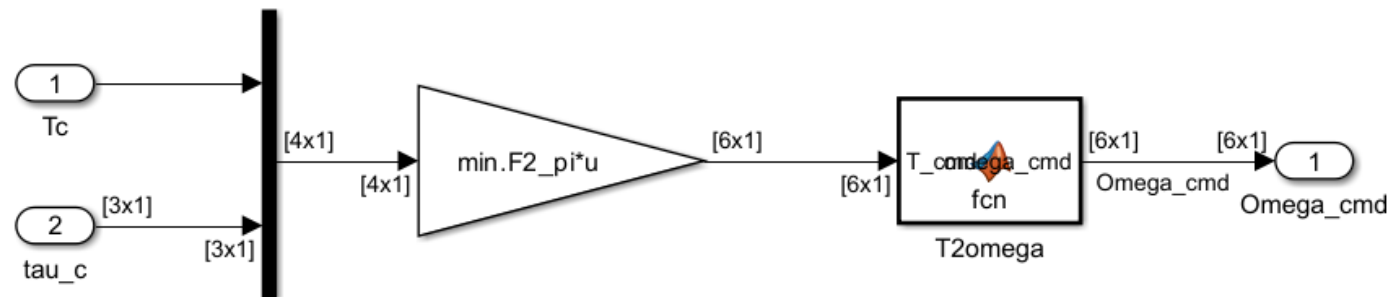


Task 2: Control allocation

$$T_c = |f_c| \quad u = \begin{bmatrix} T_c \\ \tau_{c1} \\ \tau_{c2} \\ \tau_{c3} \end{bmatrix} \quad T = \begin{bmatrix} T_1 \\ T_2 \\ T_3 \\ T_4 \\ T_5 \\ T_6 \end{bmatrix} \quad F_2 := \begin{bmatrix} 1 & \dots & 1 \\ b \sin(\gamma_1) & \dots & b \sin(\gamma_N) \\ -b \cos(\gamma_1) & \dots & -b \cos(\gamma_N) \\ -\xi_1 \sigma_1 & \dots & -\xi_N \sigma_N \end{bmatrix}$$

$$F_2^+ = F_2^\top (F_2 F_2^\top)^{-1}$$

$$T = F_2^+ u$$



Task 2: Attitude controller tuning

Requirements:

$$\omega_{\text{act}} > \omega_{\text{att}} > \omega_{\text{pos}}$$

$$T_s < 1 \text{ s}$$

$$\text{Overshoot} < 10\%$$

Tuning method:

$$R_e \approx I_3 + \alpha_e^\times$$

$$e_R \approx -\tilde{K}_R \alpha_e$$

$$\tilde{K}_R = \text{diag}([K_{R2} + K_{R3}; K_{R1} + K_{R3}; K_{R1} + K_{R2}])$$

$$\alpha_e = [\varphi_e \ \theta_e \ \psi_e]^T$$

$$J_i \ddot{\alpha}_{e_i} + K_{\omega_i} \dot{\alpha}_{e_i} + \tilde{K}_R \alpha_{e_i} = 0$$

$$K_{\omega_i} = 2\omega_i \xi_i J_i \quad \tilde{K}_{R_i} = \omega_i^2 J_i$$



Task 2: Attitude controller tuning

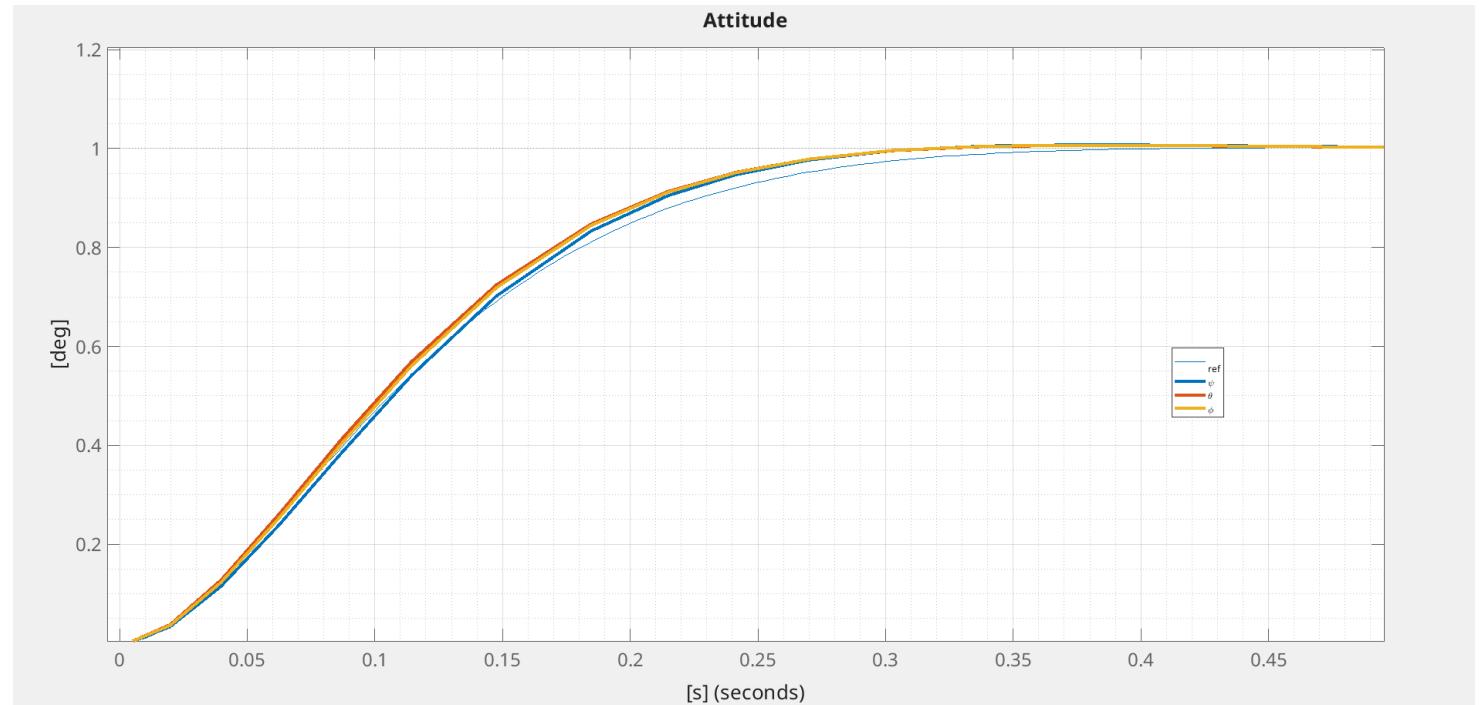
Second order system:

$$\omega = 15 \text{ rad / s} \quad \xi = 0.9$$

$$K_{\omega} = [0.27 \ 0.27 \ 0.486]$$

$$K_{R1} = 2.025 \quad K_{R2} = 2.025 \quad K_{R3} = 0.225$$

RiseTime: 0.1752
TransientTime: 0.2731
SettlingTime: 0.2731
SettlingMin: 0.1594
SettlingMax: 0.1756
Overshoot: 0.6187
Undershoot: 0
Peak: 0.1756
PeakTime: 0.3749



Task 2: Position controller tuning

Requirements:

$$\omega_{\text{pos}} < \omega_{\text{att}}$$

$$T_s < 10s$$

$$T_r < 2s$$

Tuning method:

Control System Tuner app from Simulink

Goals:

- Loop tuning: $\omega_c = 0.8 \text{ rad/s}$
- stability: $\phi_m > 45^\circ$
- Step tracking:
first order system with $\tau = 1 \text{ s}$
- Step disturbance rejection:
max amplitude = 1
 $T_s = 1 \text{ s}$
 $C_{si} = 1$



Task 2: Position controller tuning

$$K_p = 2.42 \quad K_v = 2.23 \quad K_i = 1.23$$

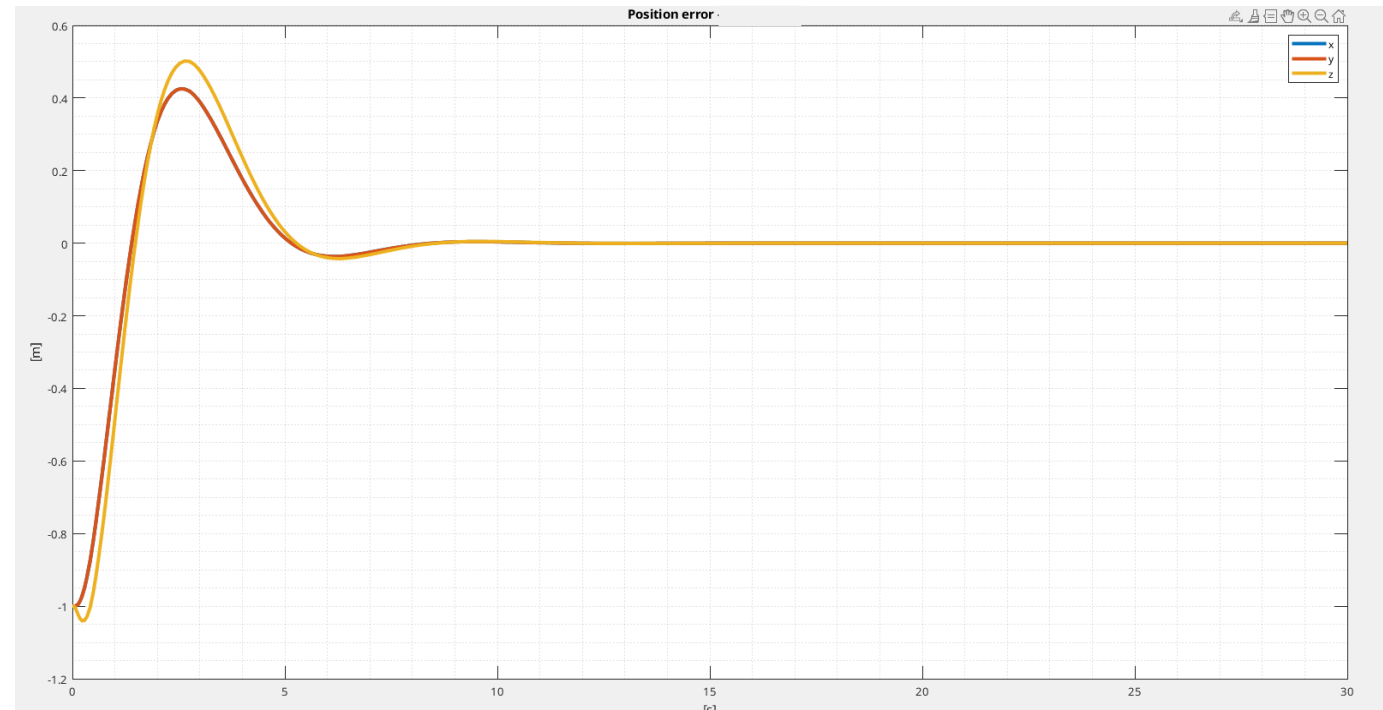
Step at [1 m ; 1m ; 1 m]

x,y

RiseTime: 0.8964
TransientTime: 7.2229
SettlingTime: 7.2229
SettlingMin: 0.9553
SettlingMax: 1.4253
Overshoot: 42.5276
Undershoot: 0
Peak: 1.4253
PeakTime: 2.5296

z

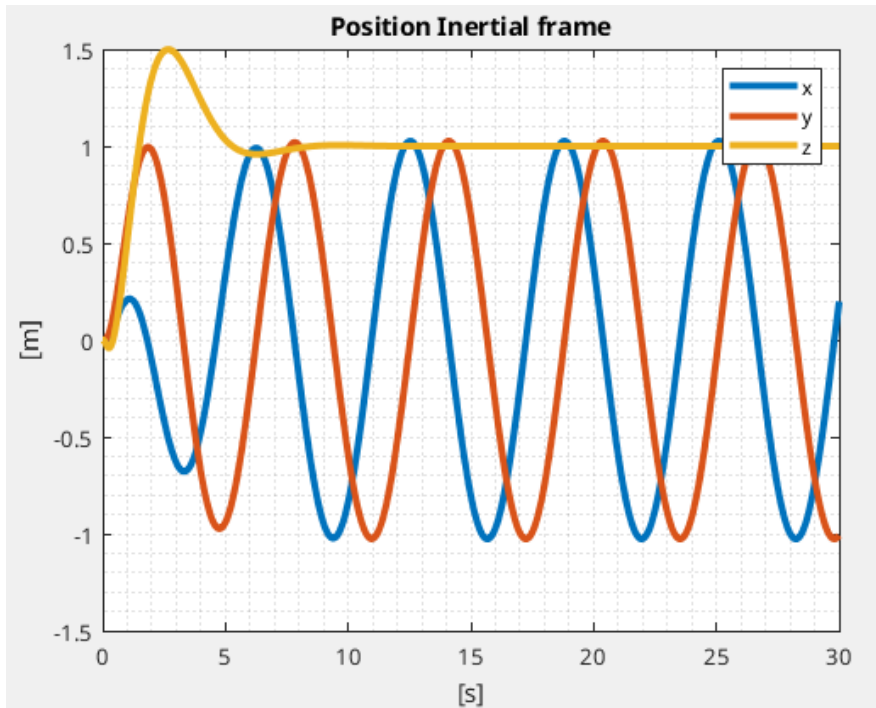
RiseTime: 0.7977
TransientTime: 7.4352
SettlingTime: 7.4684
SettlingMin: 0.9345
SettlingMax: 1.5018
Overshoot: 50.1748
Undershoot: 4.0253
Peak: 1.5018
PeakTime: 2.6356



Task 2: Position controller tuning

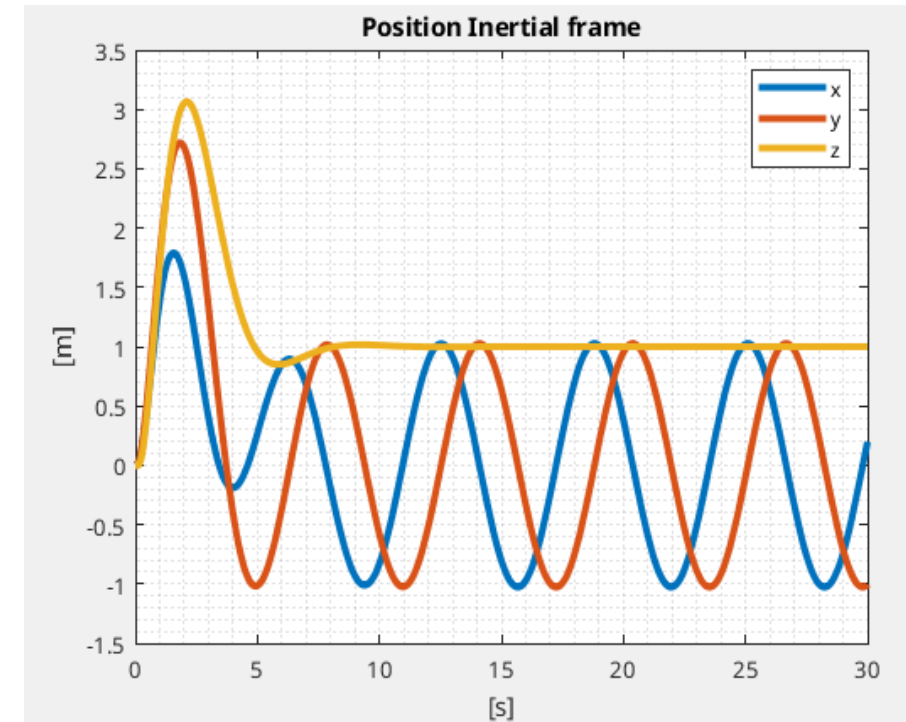
$$K_p = 2.42 \quad K_v = 2.23 \quad K_i = 1.23$$

Circle with $R = 1$ m and $\Omega = 1$ rad/s



rmse = 0.3979

Circle with $R = 1$ m and $\Omega = 1$ rad/s and
a $g * m / 2$ disturbance on each direction



rmse = 0.6195



Task 2: Full system

step

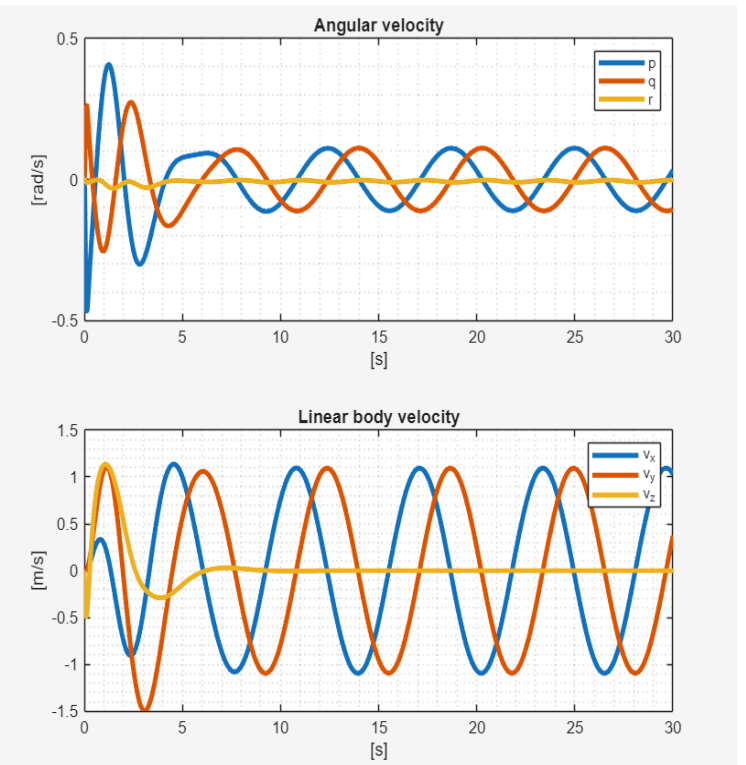
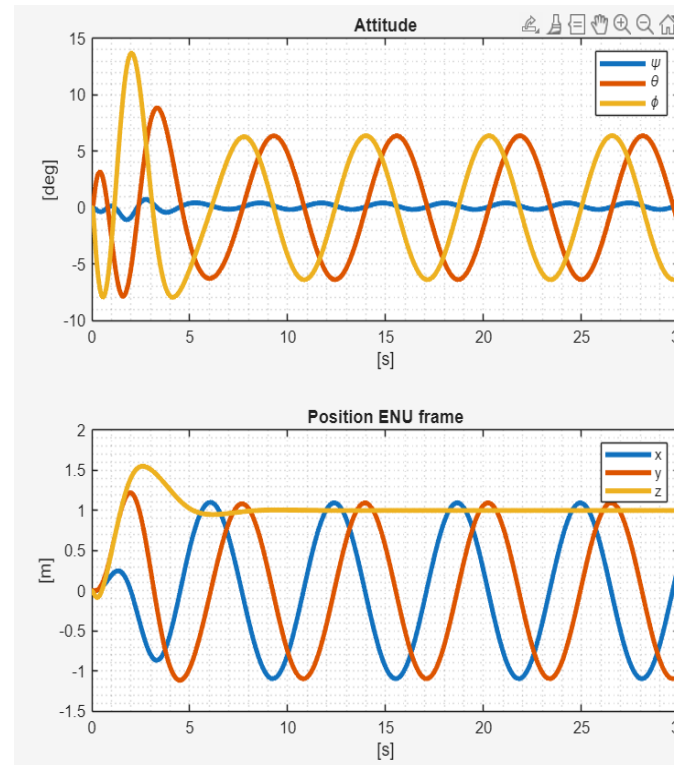
Circle with $R = 1$ m and $\Omega = 1$ rad/s

x,y

RiseTime: 0.7314
TransientTime: 5.9960
SettlingTime: 5.9960
SettlingMin: 0.9005
SettlingMax: 1.6365
Overshoot: 63.7354
Undershoot: 7.8507e-87
Peak: 1.6365
PeakTime: 2.2618

z

RiseTime: 0.7611
TransientTime: 7.5139
SettlingTime: 7.5811
SettlingMin: 0.9140
SettlingMax: 1.5301
Overshoot: 53.0089
Undershoot: 8.0640
Peak: 1.5301
PeakTime: 2.6768



rmse = 0.5568



Task 2: Retuning of attitude controller

By retuning the attitude controller for theta and phi we found faster and smoother response

$$\omega = 15 \text{ rad / s} \quad \xi = 0.1$$

$$K_{\omega} = [0.03 \ 0.03 \ 0.4]$$

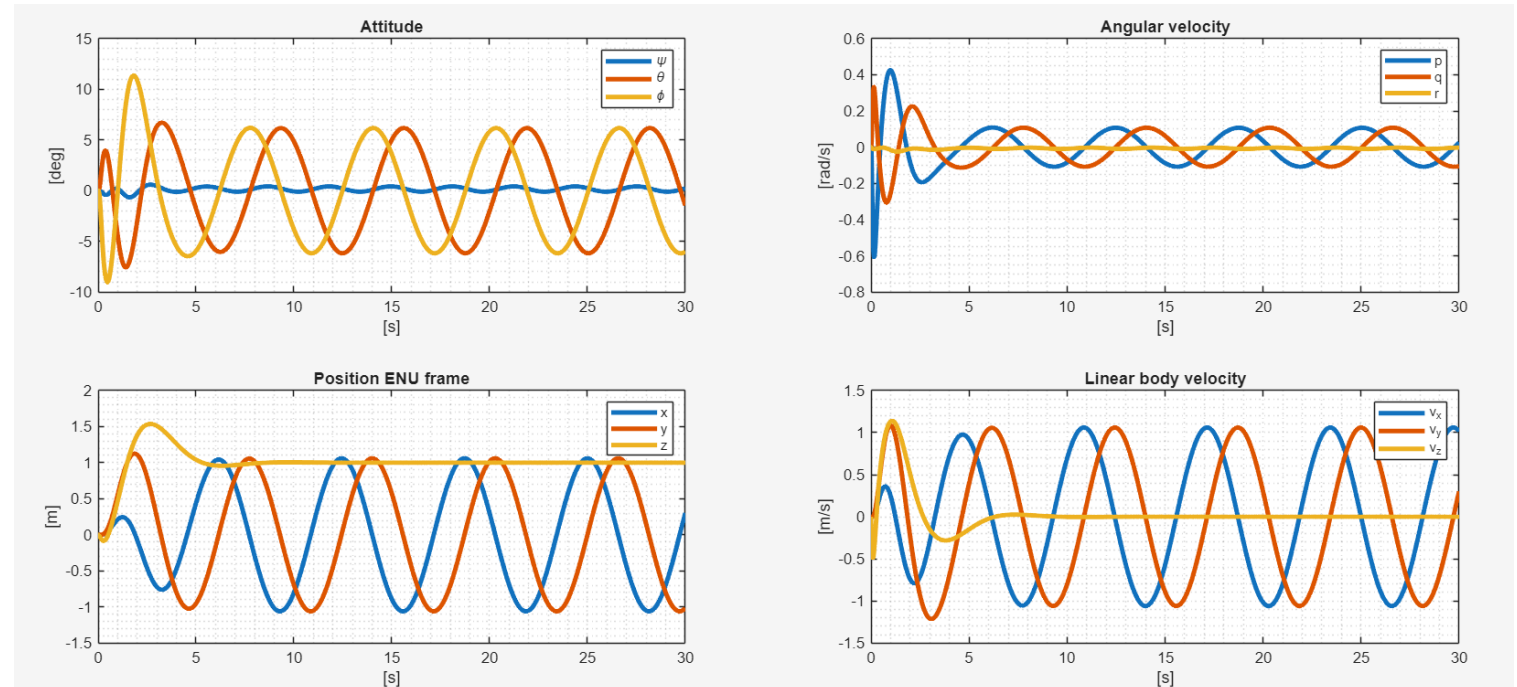
$$K_{R1} = 2.025 \quad K_{R2} = 2.025 \quad K_{R3} = 0.225$$

x,y

RiseTime: 0.7585
TransientTime: 4.9136
SettlingTime: 4.9136
SettlingMin: 0.9079
SettlingMax: 1.5026
Overshoot: 50.2557
Undershoot: 4.1017e-87
Peak: 1.5026
PeakTime: 2.3360

z

RiseTime: 0.7611
TransientTime: 7.5139
SettlingTime: 7.5811
SettlingMin: 0.9140
SettlingMax: 1.5301
Overshoot: 53.0089
Undershoot: 8.0640
Peak: 1.5301
PeakTime: 2.6768



$$\text{rmse} = 0.4957$$



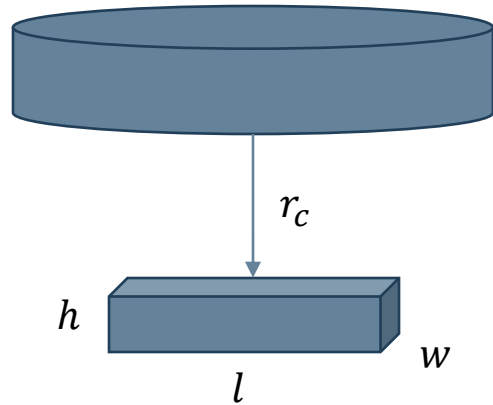
Task 2: Reduction of propeller effectiveness

	1	0.8	0.6	0.3	0
all	0.4708	0.6423	1.109	failed	/
Just one	/	0.4954	0.5389	0.7229	failed

Values of rmse for different values of propeller effectiveness considering the reduction on all propellers or on just one.



Task 3: package delivery



Dimensions:

- Mass = 0.5 Kg
- Height $[h] = 0.03$ m
- Length $[l] = 0.2$ m
- Width $[w] = 0.1$ m
- Position $[rc] = [0, 0, -0.1]$ m

Inertial properties:

- $I_{xx} = 1/12m(h^2 + l^2) + m(rc_y^2 + rc_z^2)$
- $I_{yy} = 1/12m(h^2 + w^2) + m(rc_x^2 + rc_z^2)$
- $I_{zz} = 1/12m(w^2 + l^2)$

Task 3: package delivery

Sinusoidal trajectory: $x(t) = A \sin(\omega t)$, $A = 1$, $\omega = 1$, $z_d = 1$, $y_d = 0.5 t$.

Root mean square error:

$$rmse1 = \sqrt{\text{mean} \left(\sqrt{(x - x_d)^2 + (y - y_d)^2 + (z - z_d)^2} \right)}$$

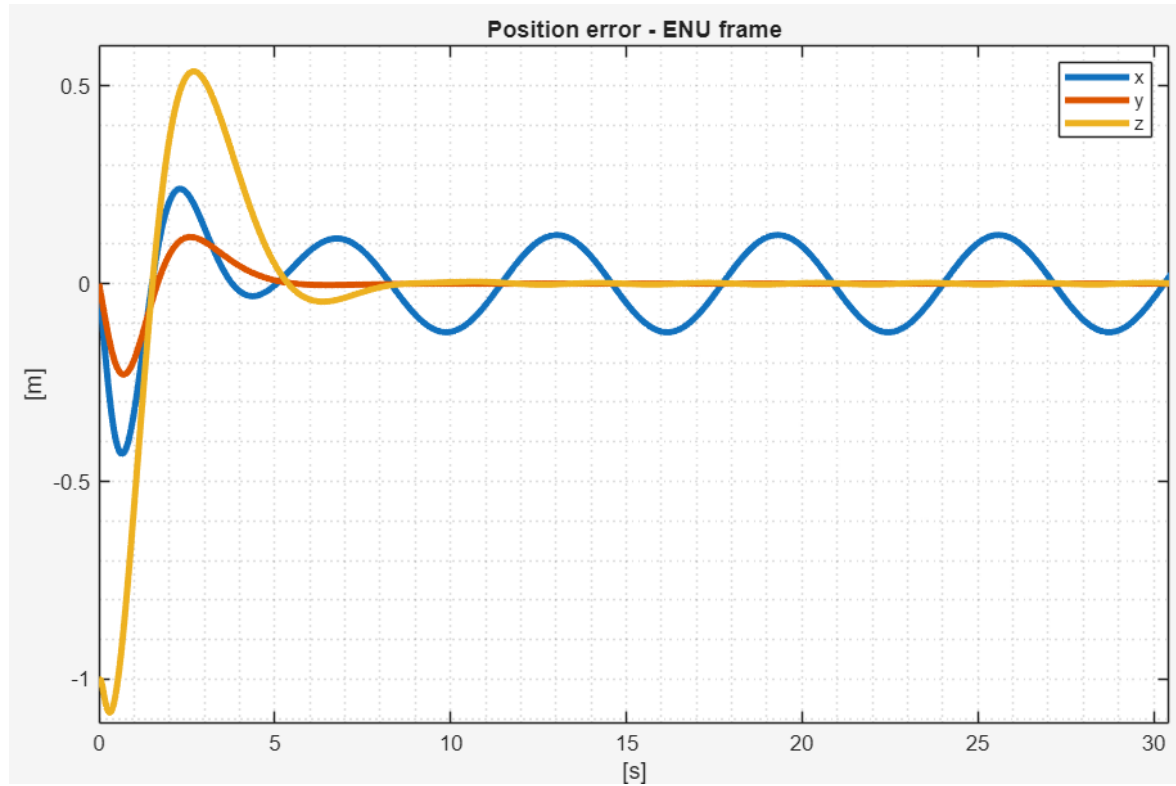
rmse2 is evaluated analogously but considering the positions x , y , z starting from 10 seconds of simulation.

Setting ($T_s = 1000 \text{ s}$)	rmse1	rmse2
Baseline controller without package	0.3167	0.3128
Baseline controller with package	0.4078	0.4044

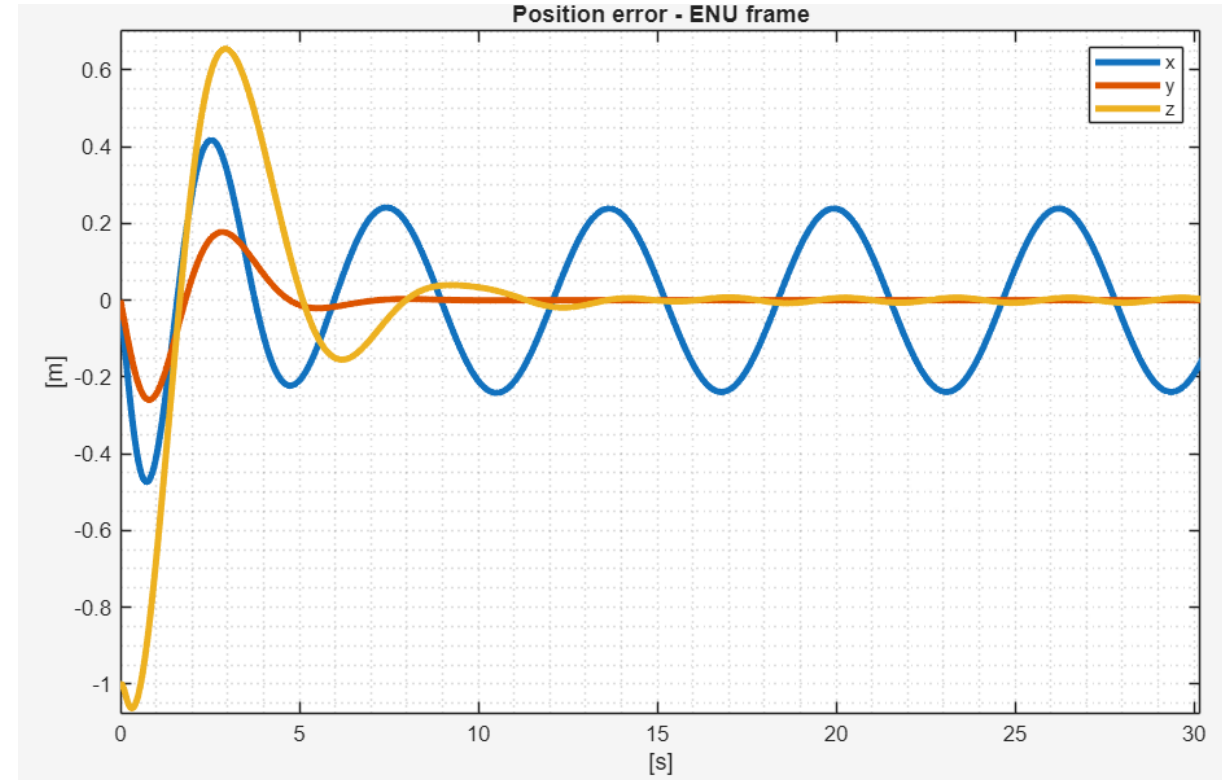


Task 3: package delivery

Baseline without package



Baseline with package



Task 3: Adaptive augmentation – PB MRAC dynamics

UAV dynamics:

$$\begin{cases} \dot{p} = v \\ m\dot{v} = -gme_3 + f_e + \Lambda_f f_c \end{cases}$$

Adaptive laws:

$$\begin{cases} \dot{\hat{p}} = \hat{v} \\ \dot{\hat{v}} = 1/m(\hat{\Lambda}_f u + \varphi_p^T \hat{\theta}_p) + L(v - \hat{v}) - ge_3 \end{cases}$$

$$\begin{cases} \dot{\hat{x}}_p = A_p \hat{x}_p + B_p(\hat{\Lambda}_f u + \varphi_p^T \hat{\theta}_p) + L\hat{e}_p - [0; 0; 0; ge_3] \\ u_{c,AD} = (\hat{\Lambda}_f^{-1} - I)u_{c,BL} - \hat{\Lambda}_f^{-1} \varphi_p^T \hat{\theta}_p \\ \dot{\hat{\theta}}_a = \Gamma_a \varphi_a B_p^T \hat{e}_p \end{cases}$$

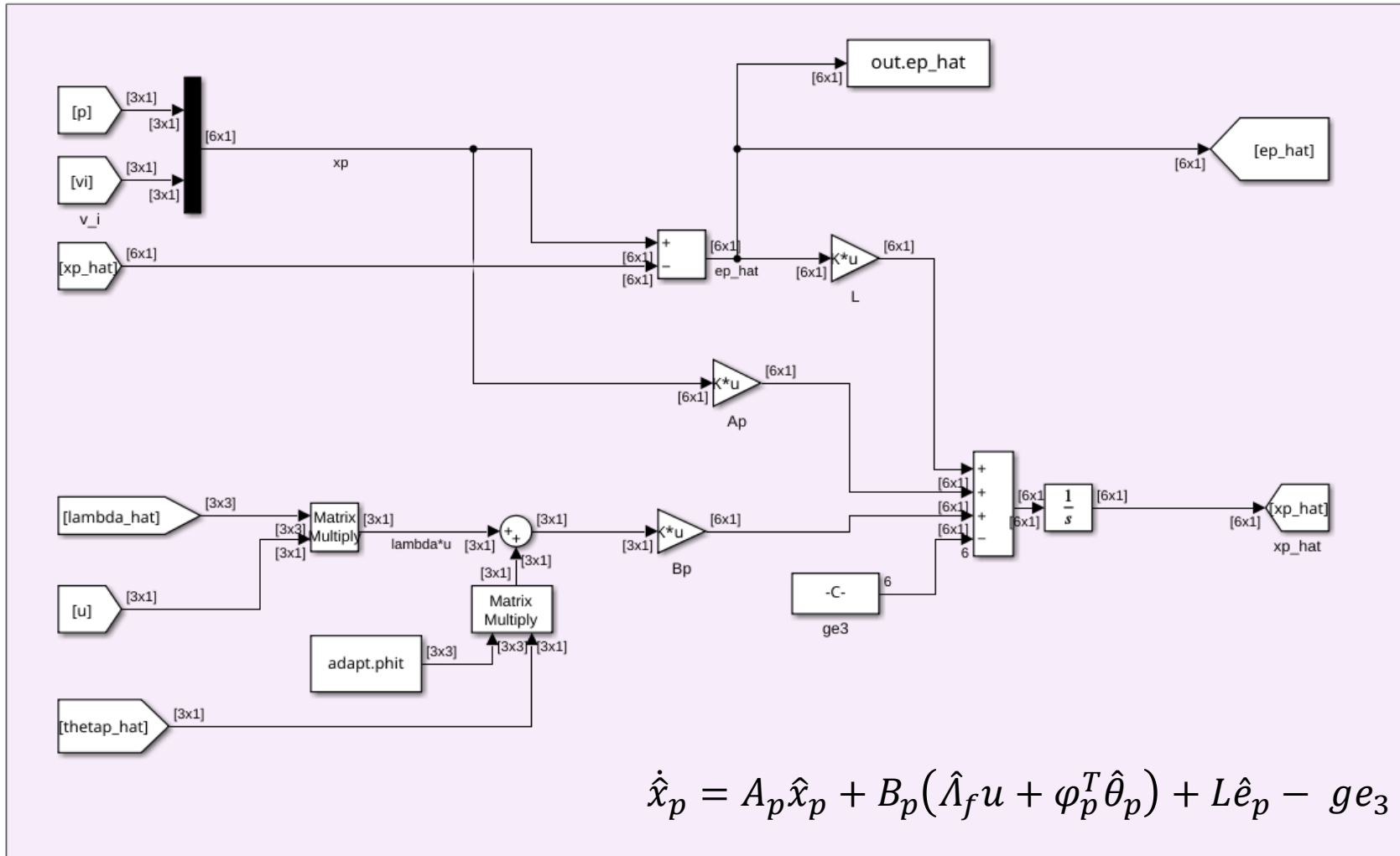
$$A_p = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix},$$

$$B_p = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1/m & 0 & 0 \\ 0 & 1/m & 0 \\ 0 & 0 & 1/m \end{bmatrix}$$

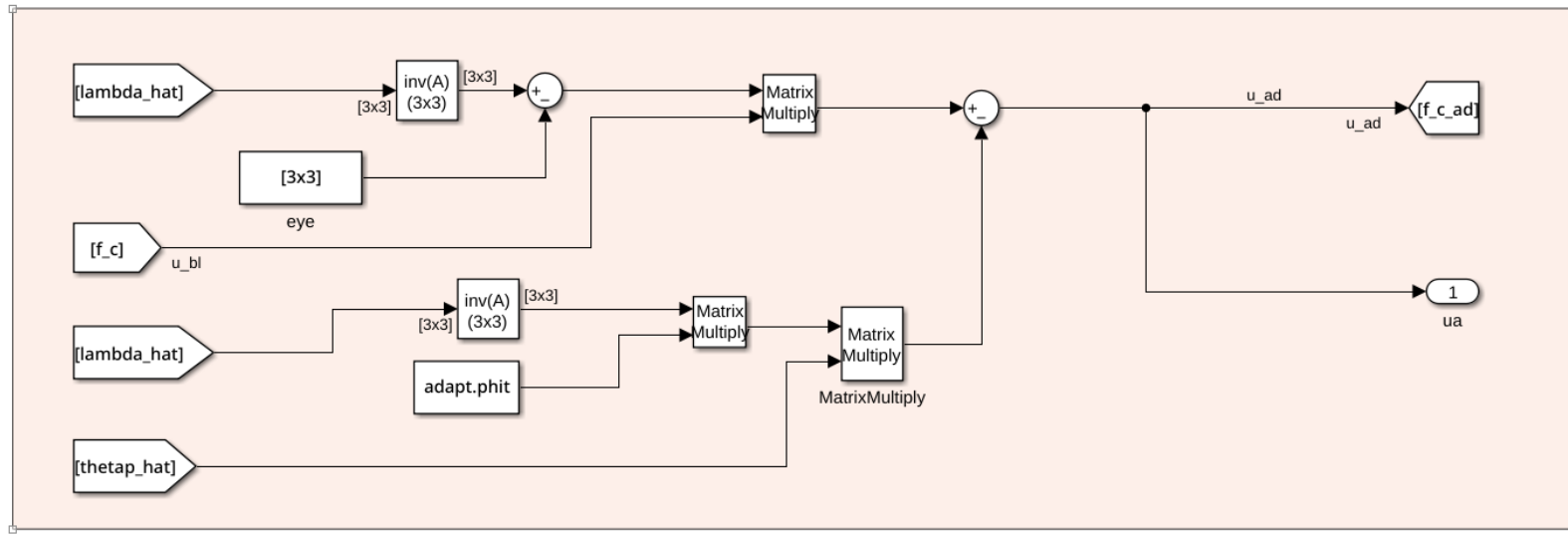
Where $\hat{x}_p = [\hat{p}, \hat{v}]$, $\hat{e}_p = x_p - \hat{x}_p$, $\varphi^T = I$



Task 3: Adaptive augmentation – Simulink implementation

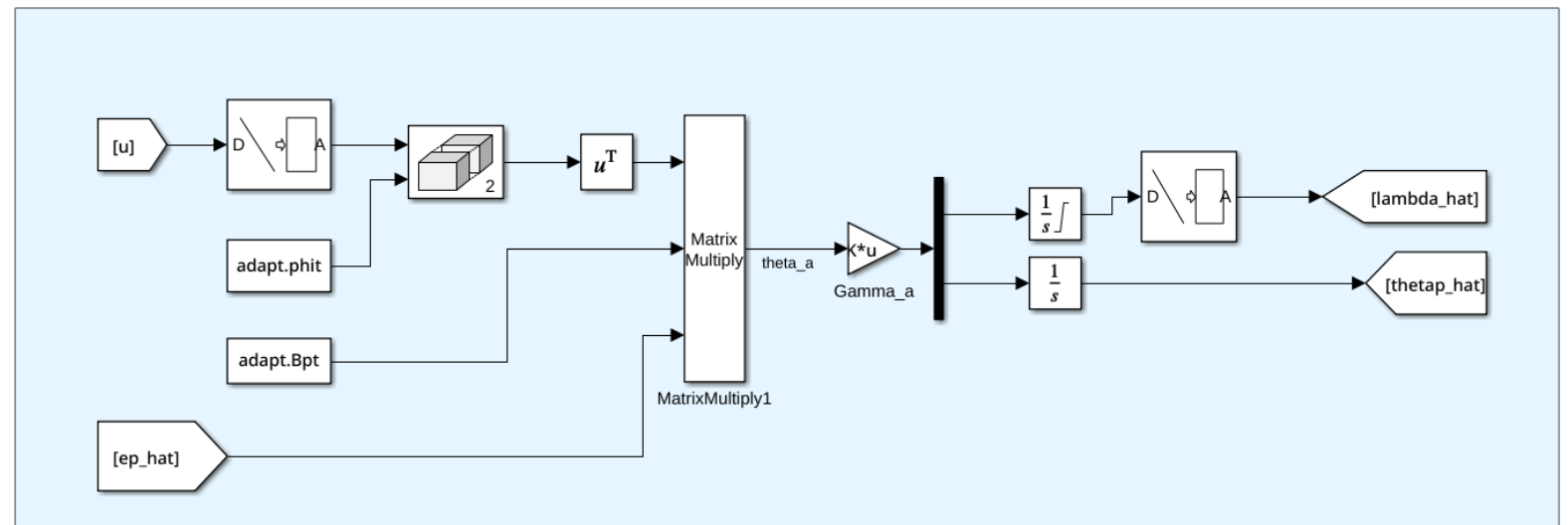


Task 3: Adaptive augmentation – Simulink implementation



$$u_{c,AD} = (\hat{\Lambda}_f^{-1} - I)u_{c,BL} - \hat{\Lambda}_f^{-1}\varphi_p^T\hat{\theta}_p$$

$$\dot{\hat{\theta}}_a = \Gamma_a \varphi_a B_p^T \hat{e}_p$$



Task 3: Adaptive augmentation – Tuning of the adaptive gains

Saturation introduced in the integration of $\dot{\hat{\theta}}_a$ for the derivation of $\hat{\Lambda}_f$, ensuring feasible matrix inversion: $\hat{\Lambda}_f = 0.1 \div 1.3$.

Adaptive gains:

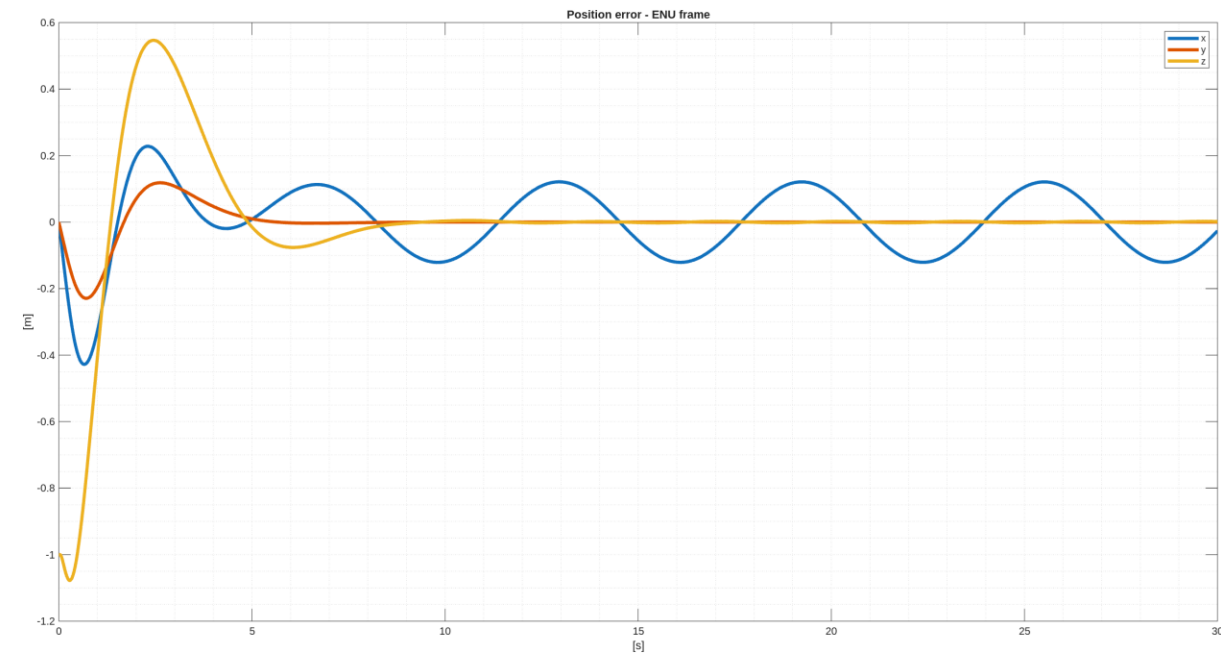
- $L = 15 \mathbf{I}_{6 \times 6}$
- $\Gamma = 0.1 \mathbf{I}_{6 \times 6}$

Setting ($T_s = 1000 \text{ s}$)	rmse1	rmse2
Baseline controller with package	0.4078	0.4044
Augmented controller with package	0.3423	0.3377
Baseline controller without package	0.3167	0.3128
Augmented controller without package	0.3144	0.3171

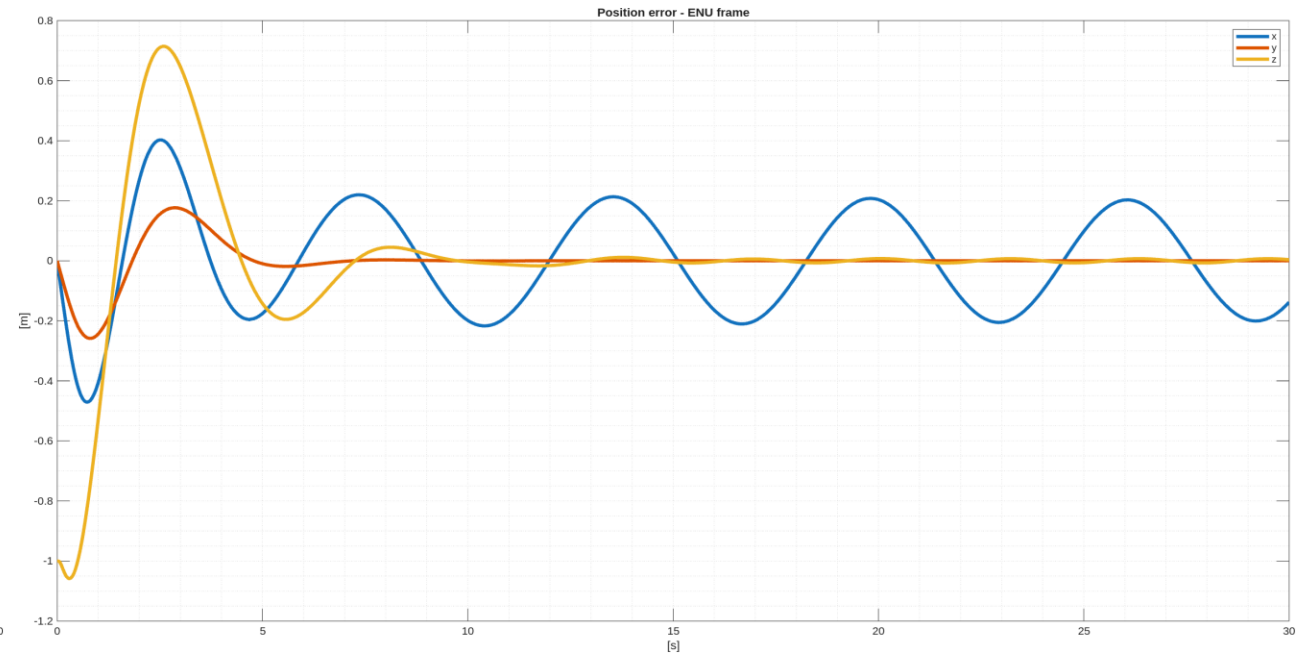


Task 3: Adaptive augmentation – Trajectory tracking results for PB-MRAC

PB-MRAC without package

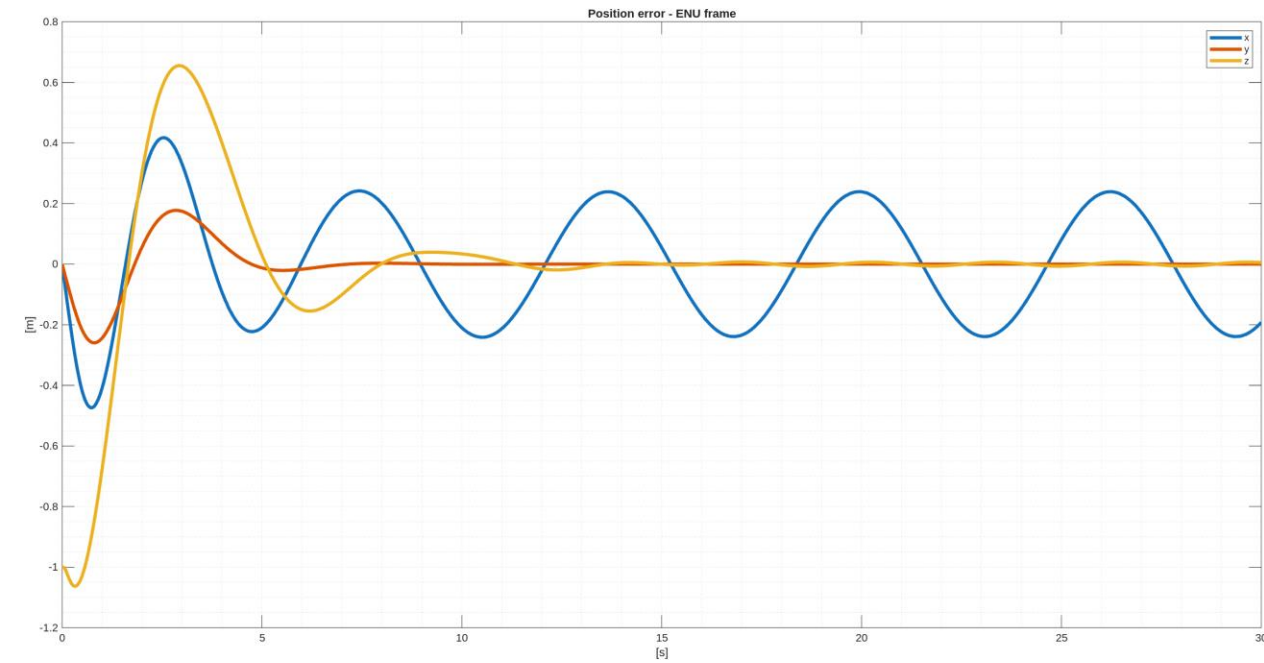


PB-MRAC with package

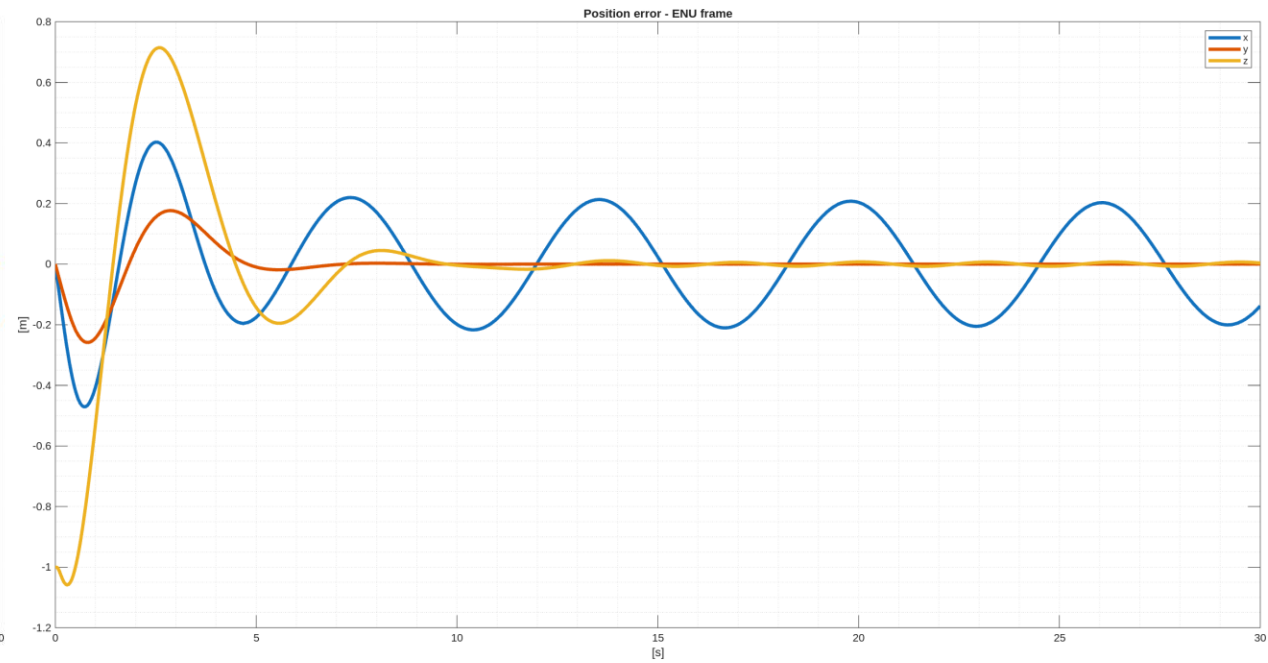


Task 3: Adaptive augmentation – Trajectory tracking results for PB-MRAC

Baseline with package

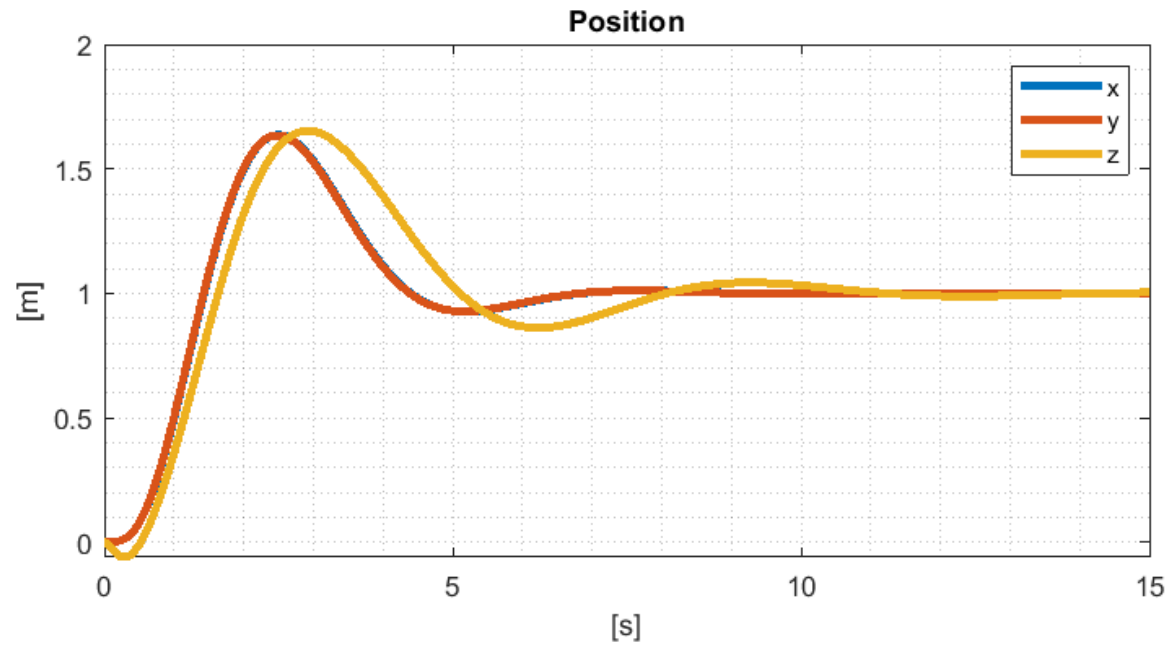


PB-MRAC with package

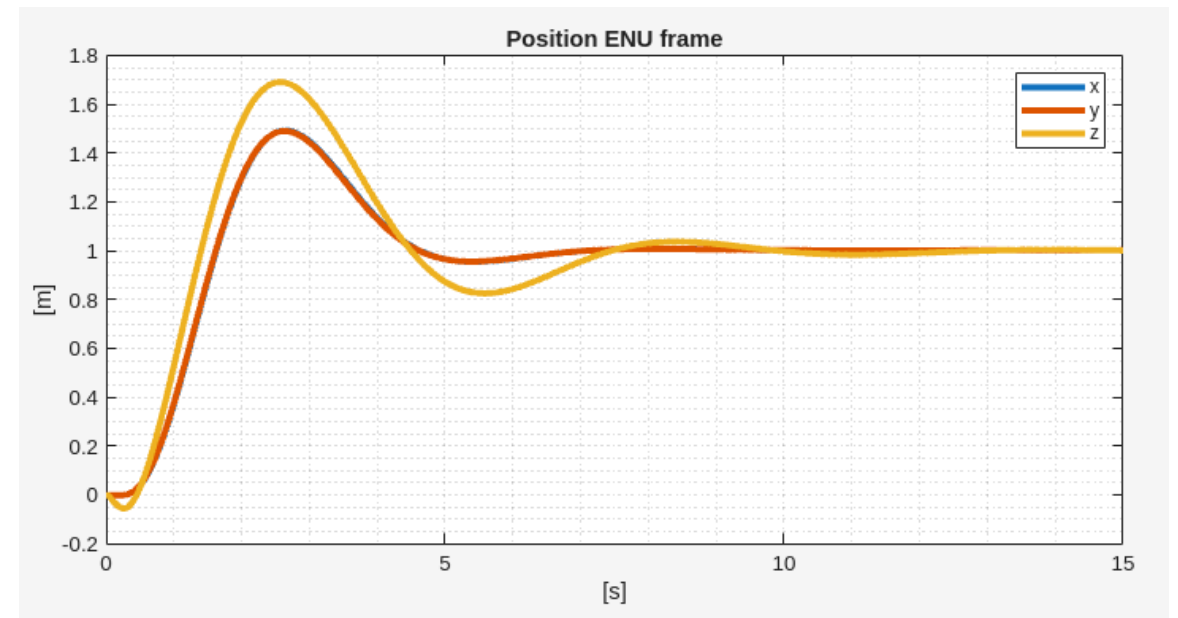


Task 3: Adaptive augmentation – set point tracking

Baseline with package



PB-MRAC with package



Task 3: Adaptive augmentation – Addition of a disturbance in PB-MRAC

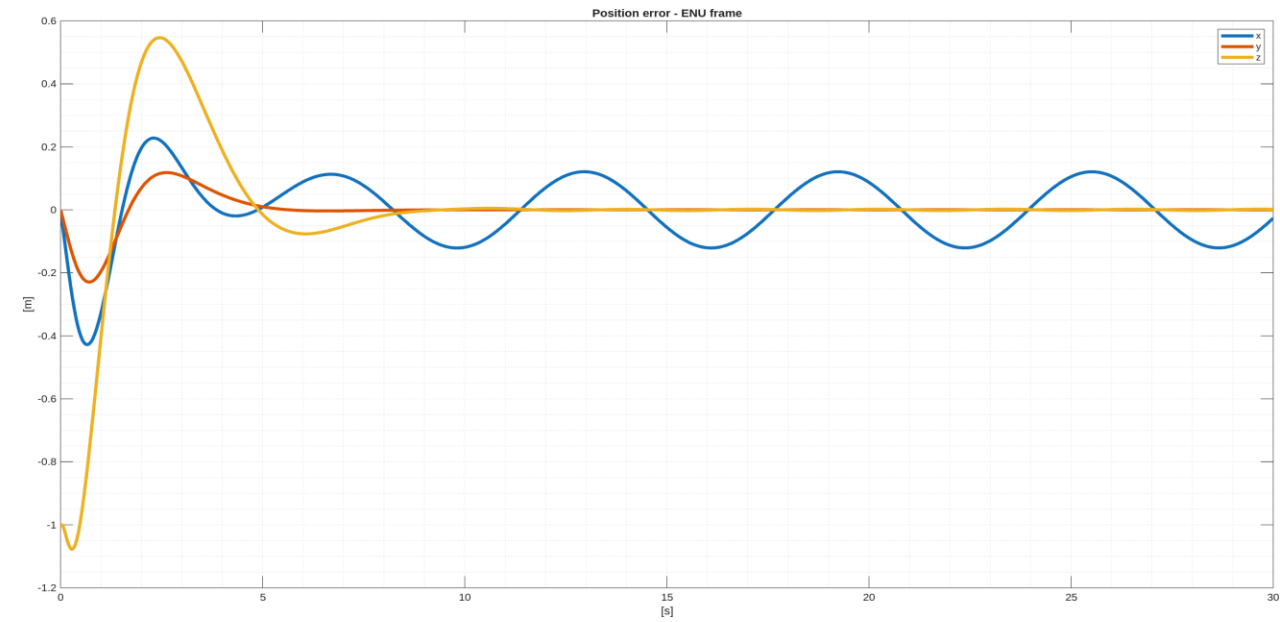
Simulation of the sinusoidal trajectory. The wind gust disturbance considered is $v_w = [10, 10, 0]$ m/s and is applied as a step at 0 seconds.

Setting ($T_s = 1000$ s)	rmse1	rmse2
With package and no disturbance	0.3516	0.3377
With package and wind gust disturbance	0.3488	0.3366
Without package and no disturbance	0.3144	0.3107
Without package and wind gust disturbance	0.3212	0.3174

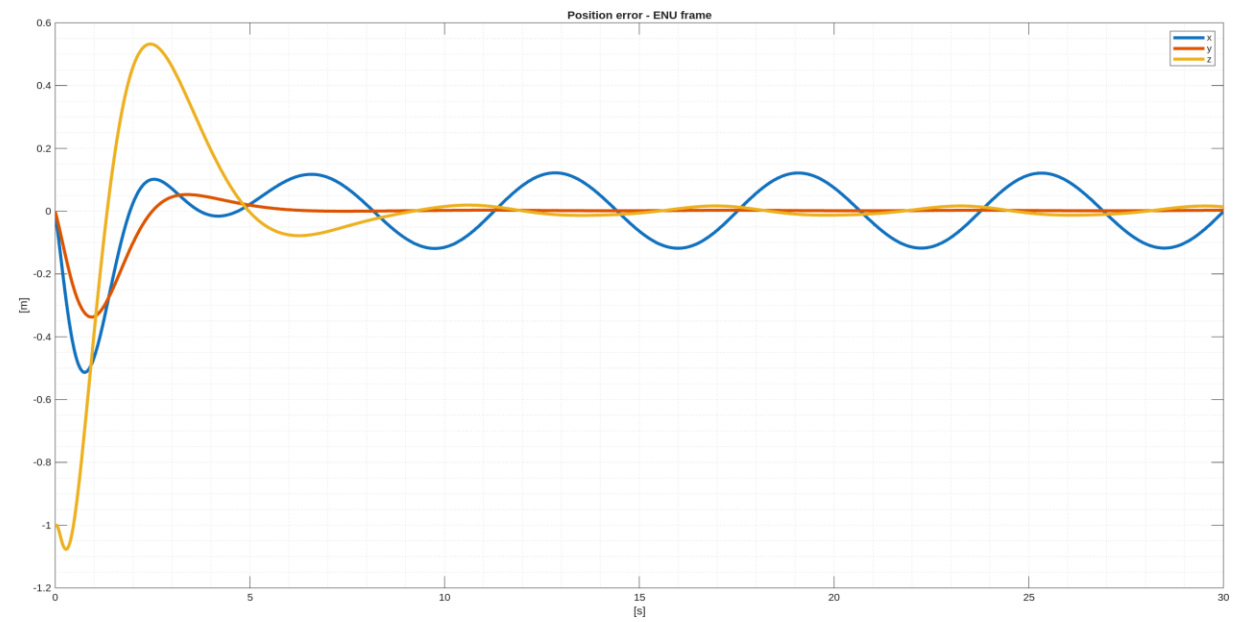


Task 3: Adaptive augmentation – Addition of a disturbance

PB-MRAC without package and no disturbance

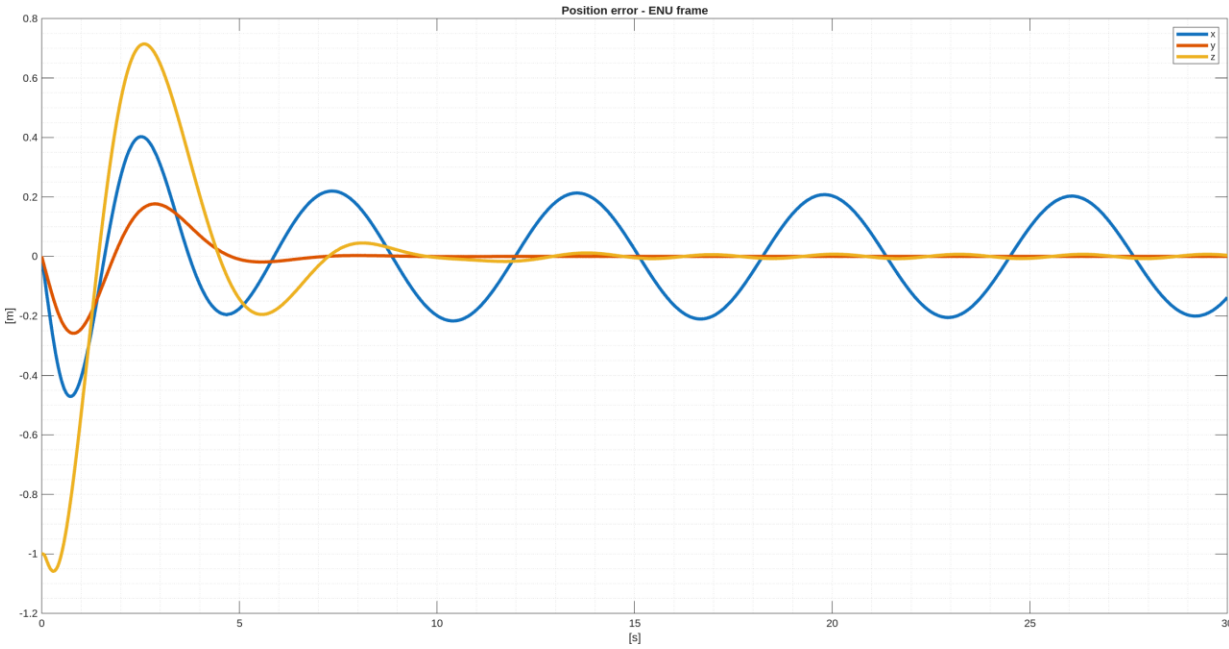


PB-MRAC without package and additional disturbance

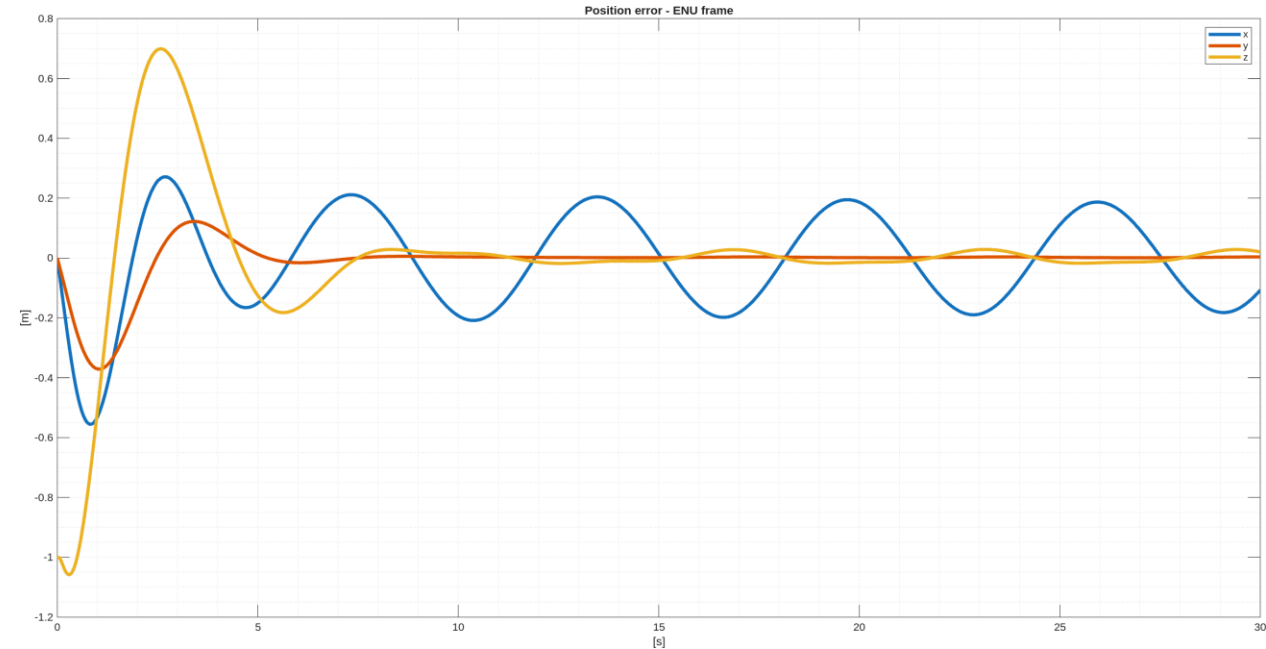


Task 3: Adaptive augmentation – Addition of a disturbance

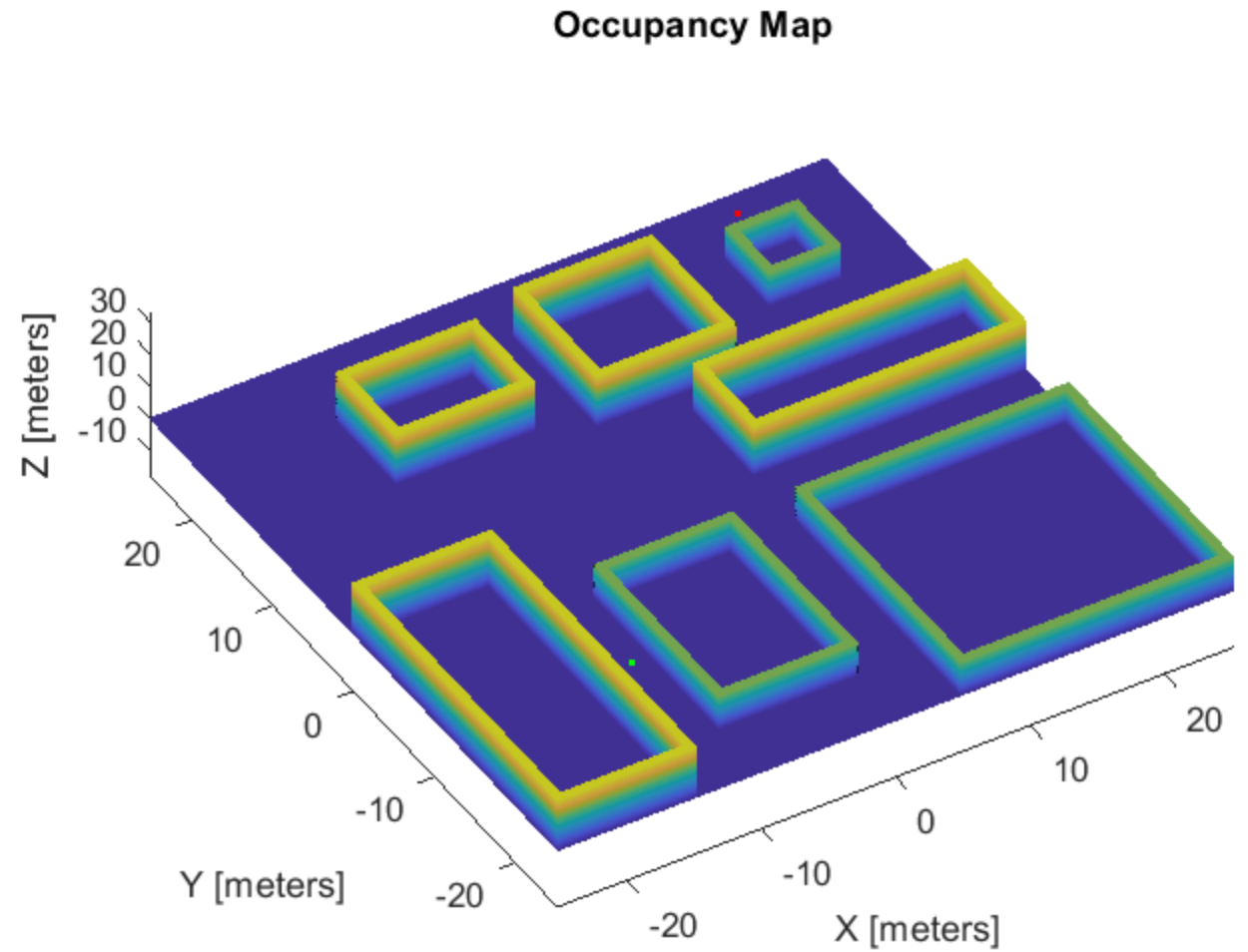
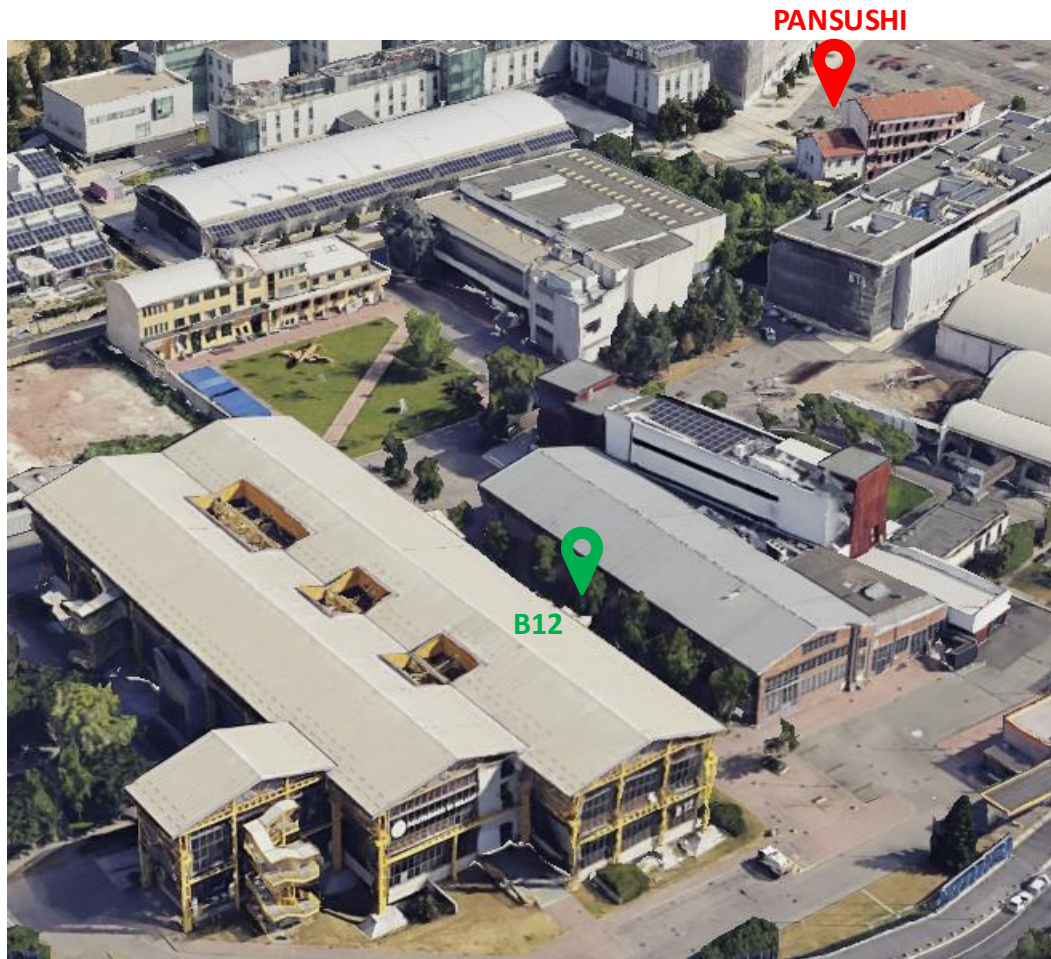
PB-MRAC with package and no disturbance



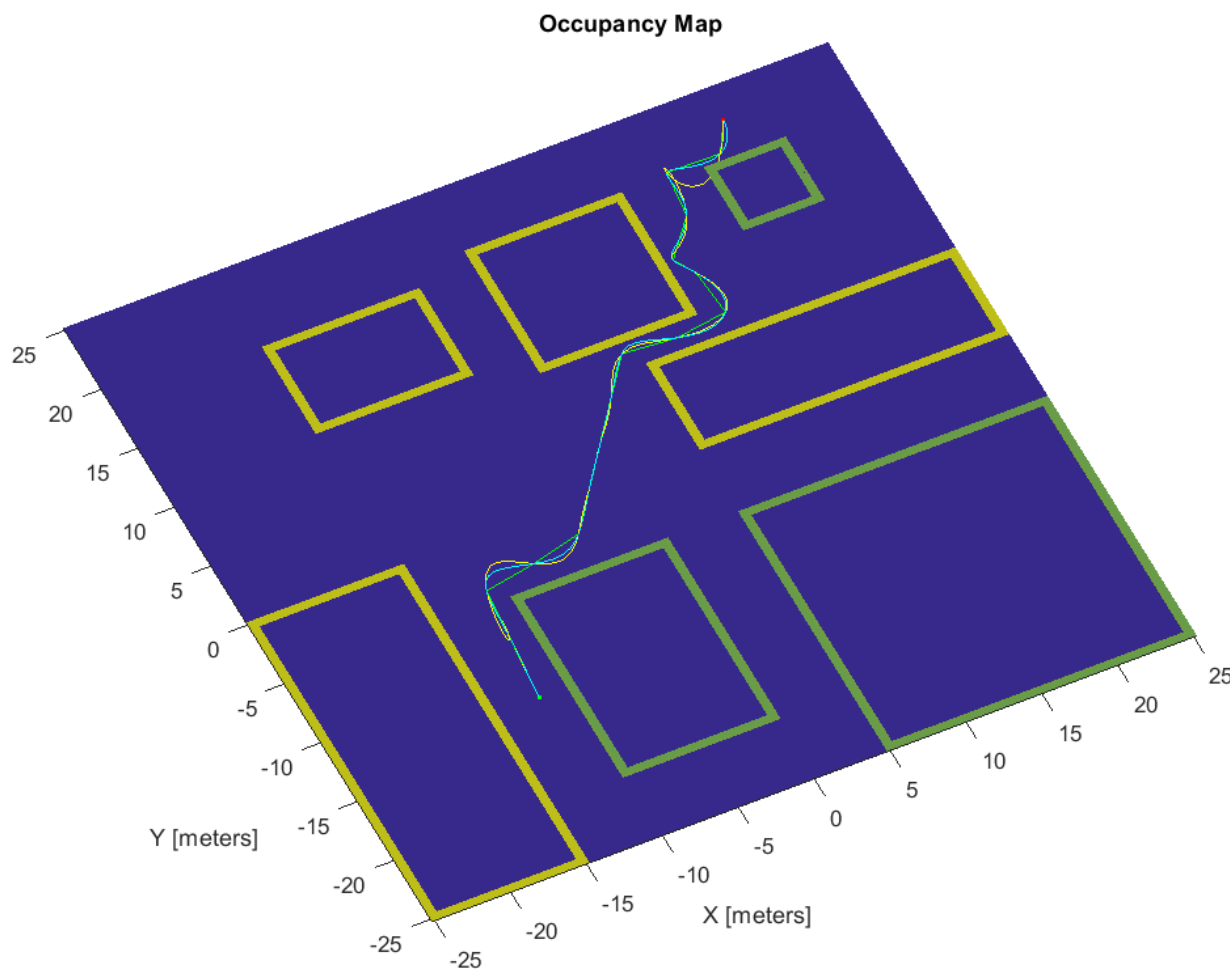
PB-MRAC with package and additional disturbance



Task 4: Test in Delivery Scenario



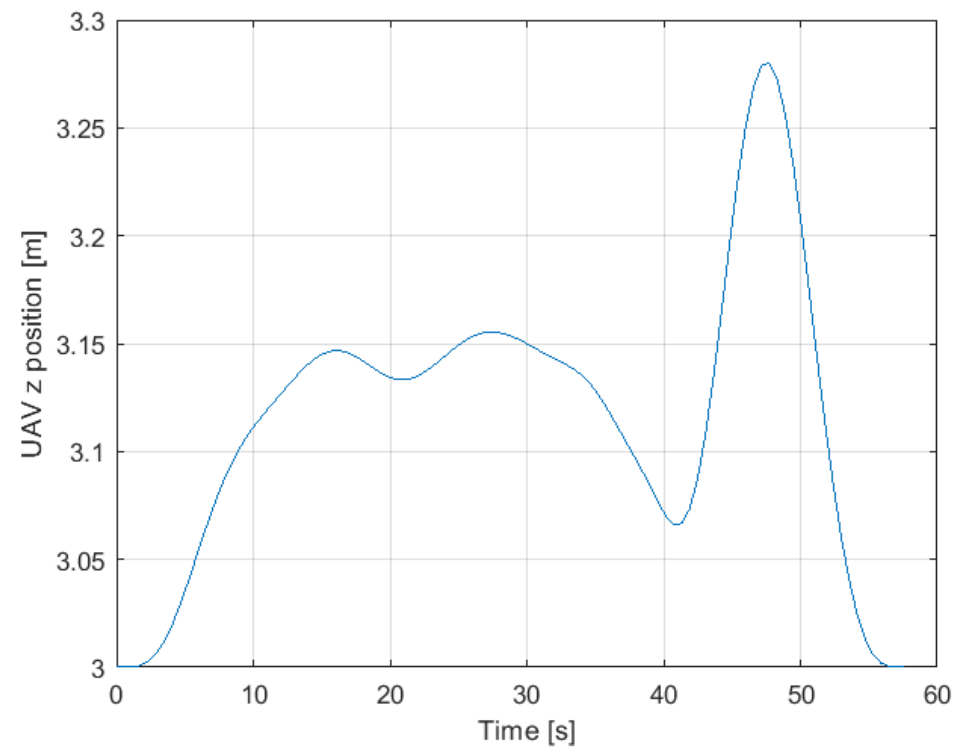
Task 4: Test in Delivery Scenario



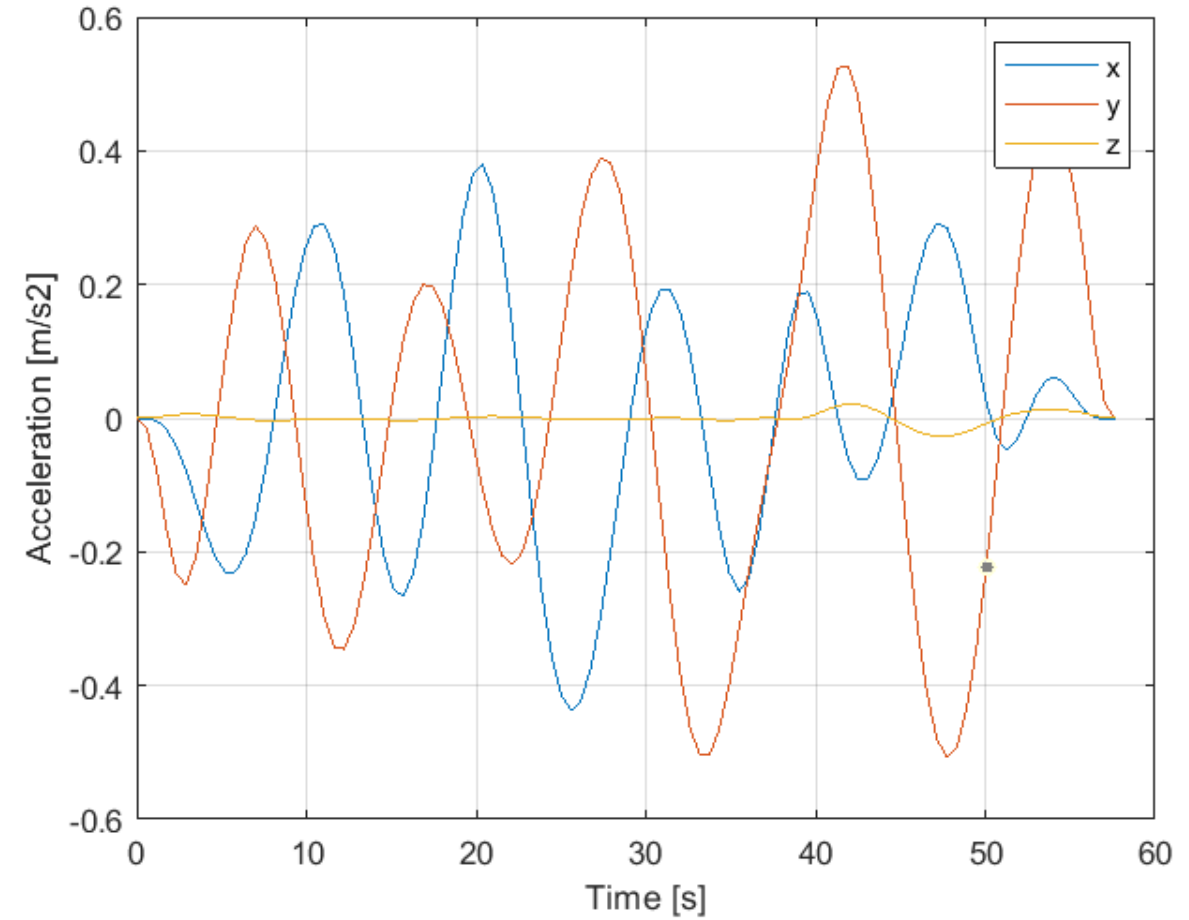
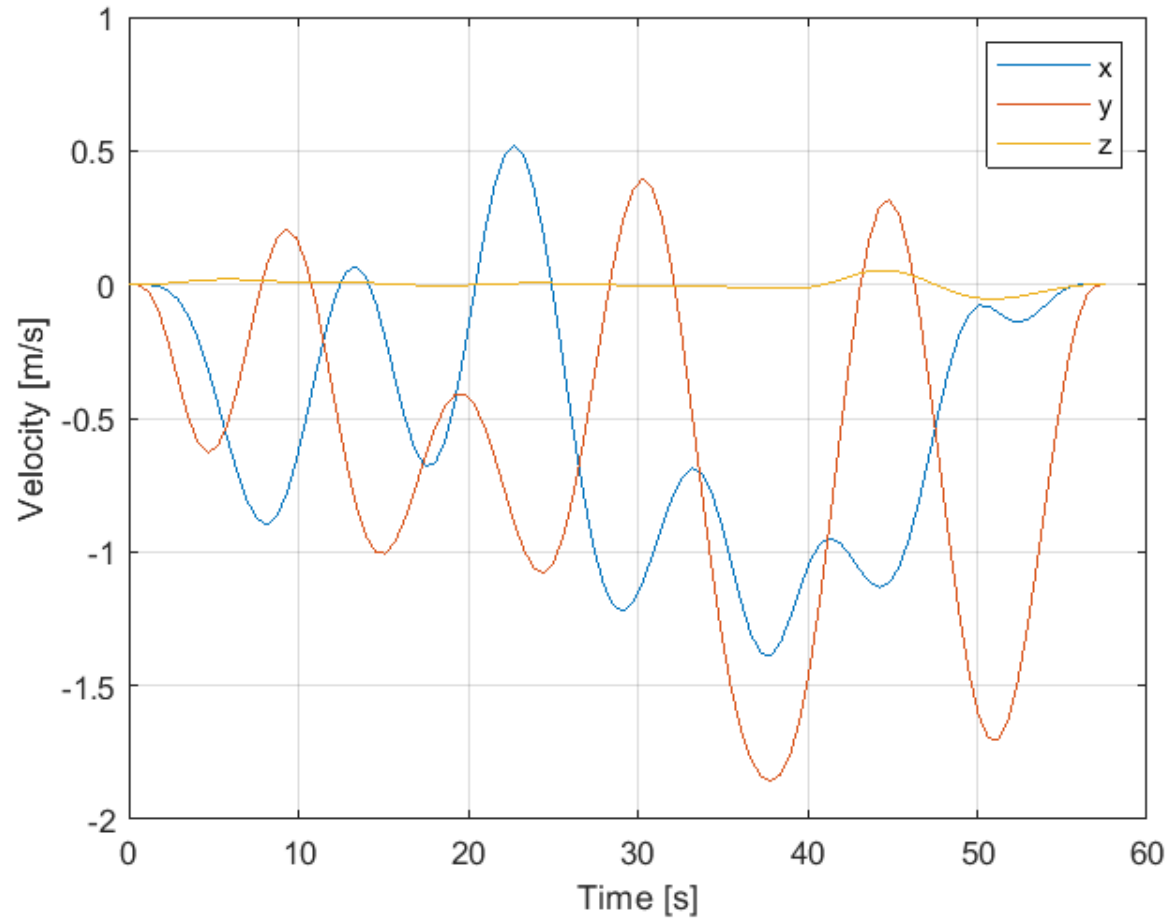
- Start Position
- Goal Position
- Planned Path
- Initial Trajectory
- Valid Trajectory

planner.MaxIterations = 1000;
planner.MaxNumTreeNode = 1000;
TimeWeight=0.1;

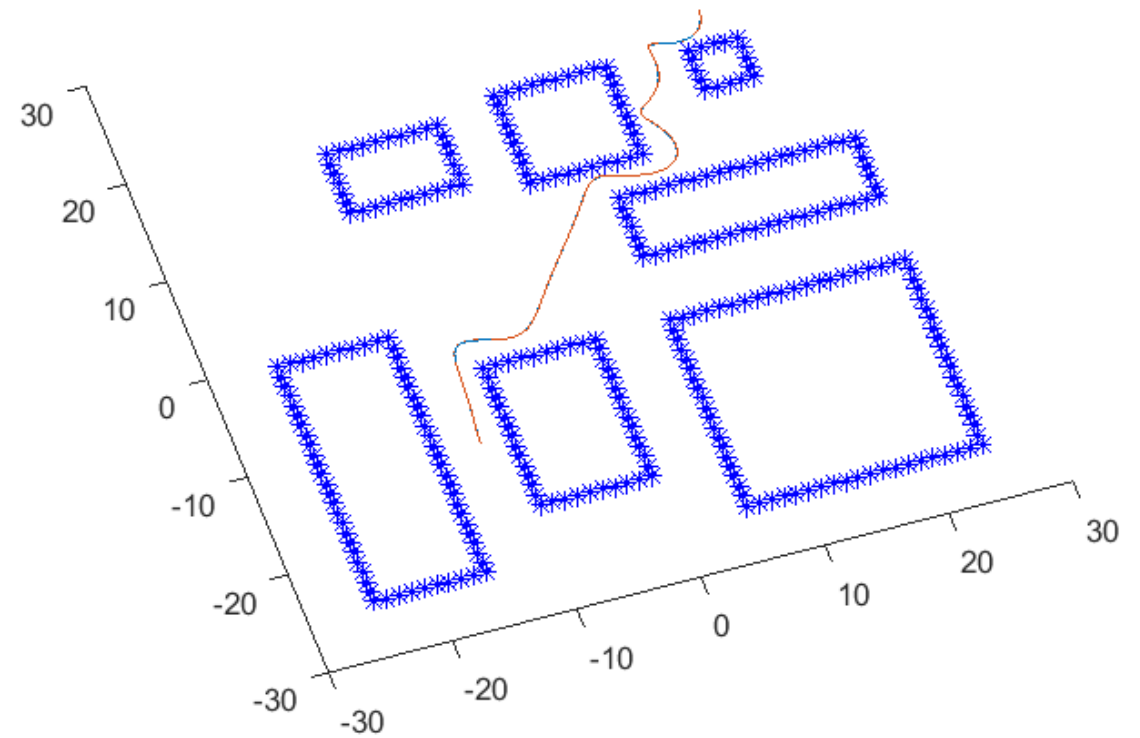
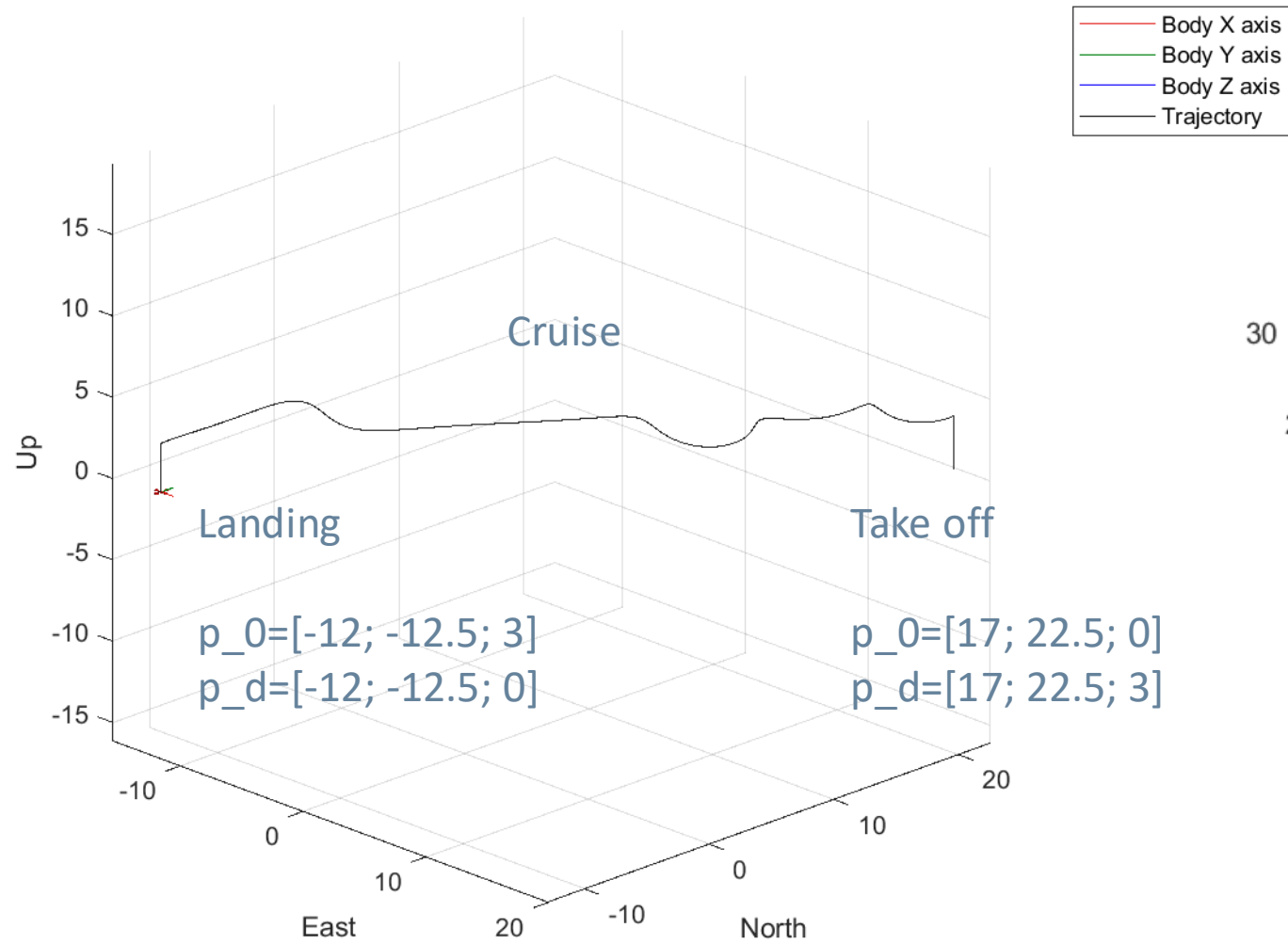
$z=[3:3.5];$ [m]



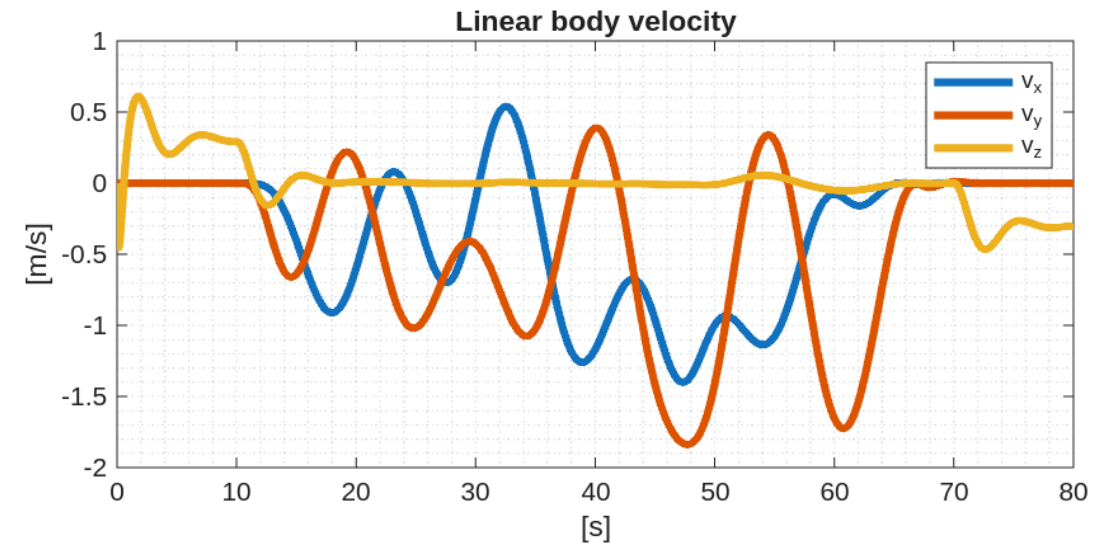
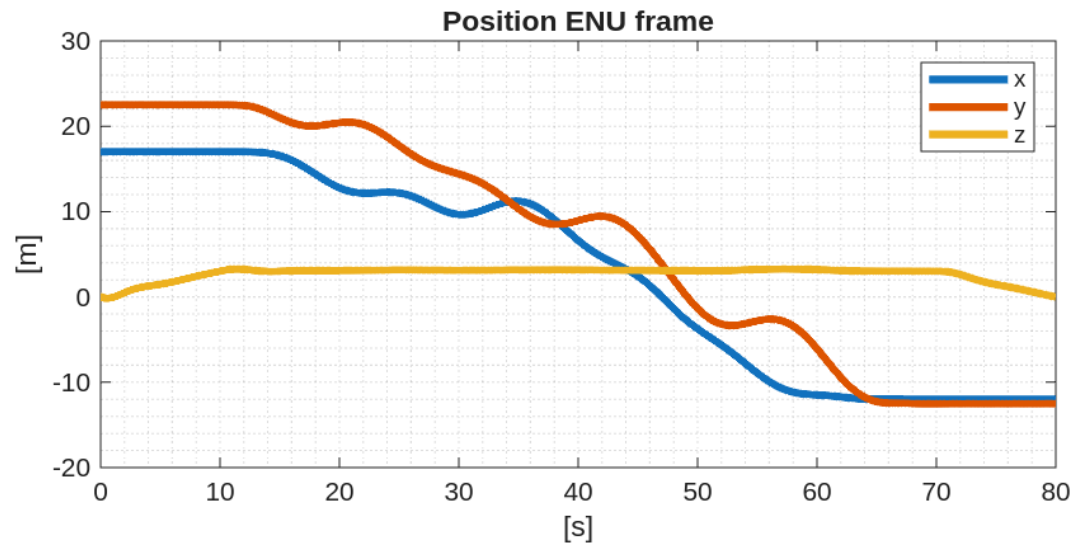
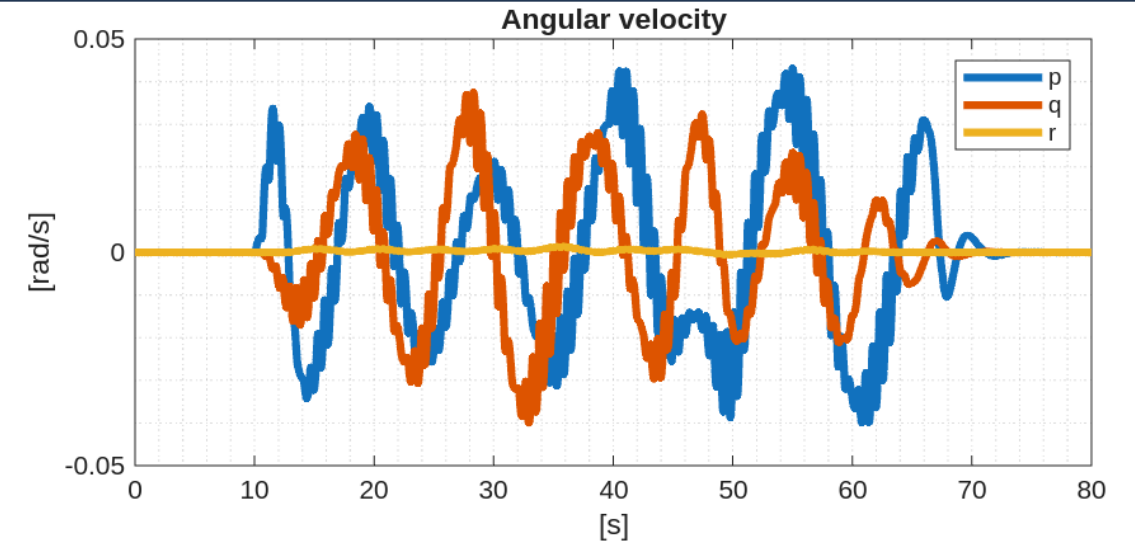
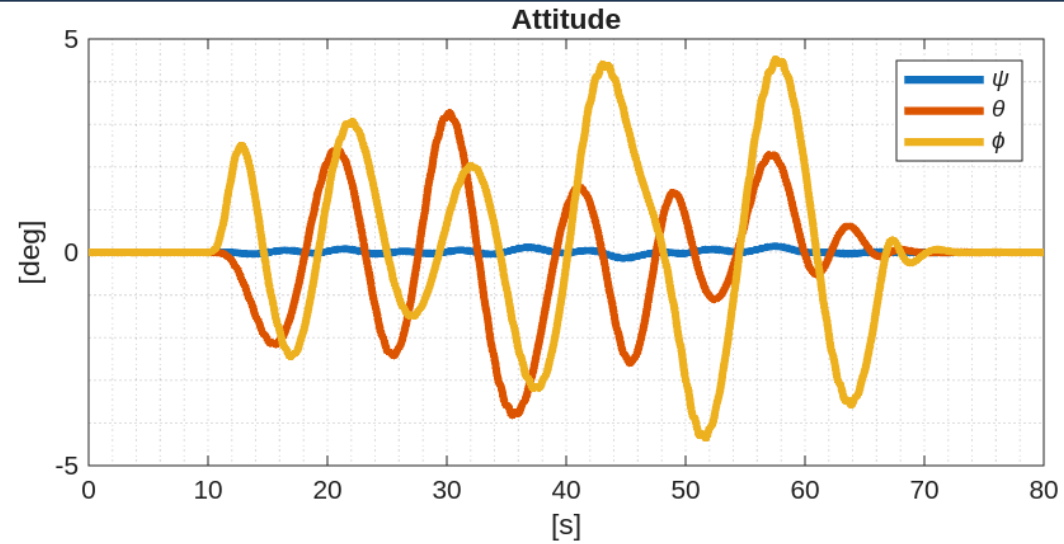
Task 4: Test in Delivery Scenario



Task 4: Test in Delivery Scenario

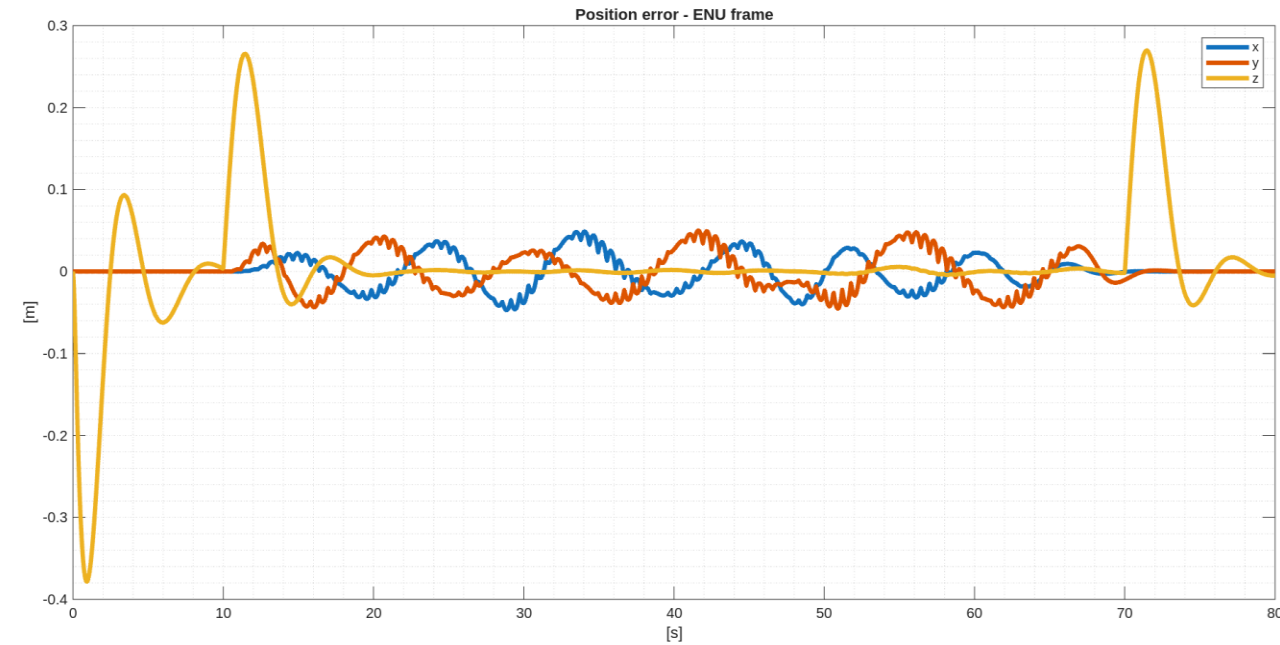


Task 4: Test in Delivery Scenario

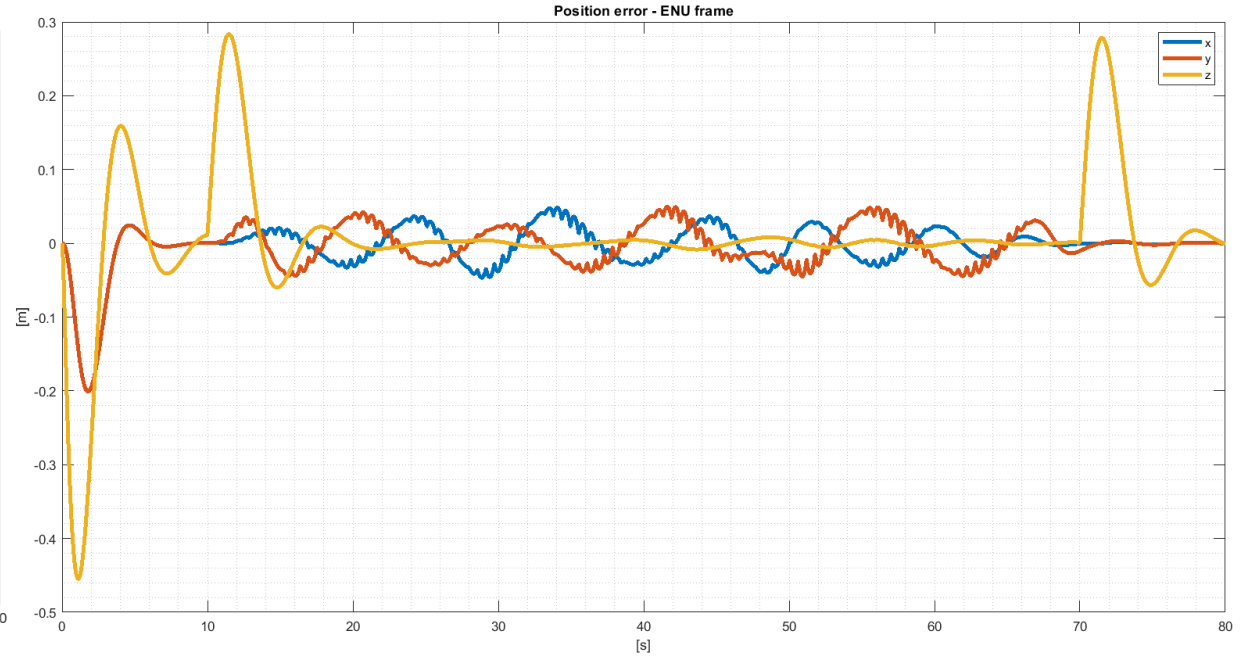


Task 4: Test in Delivery Scenario

Wind=[10; 10; 0] [m/s]



With PB-MRAC

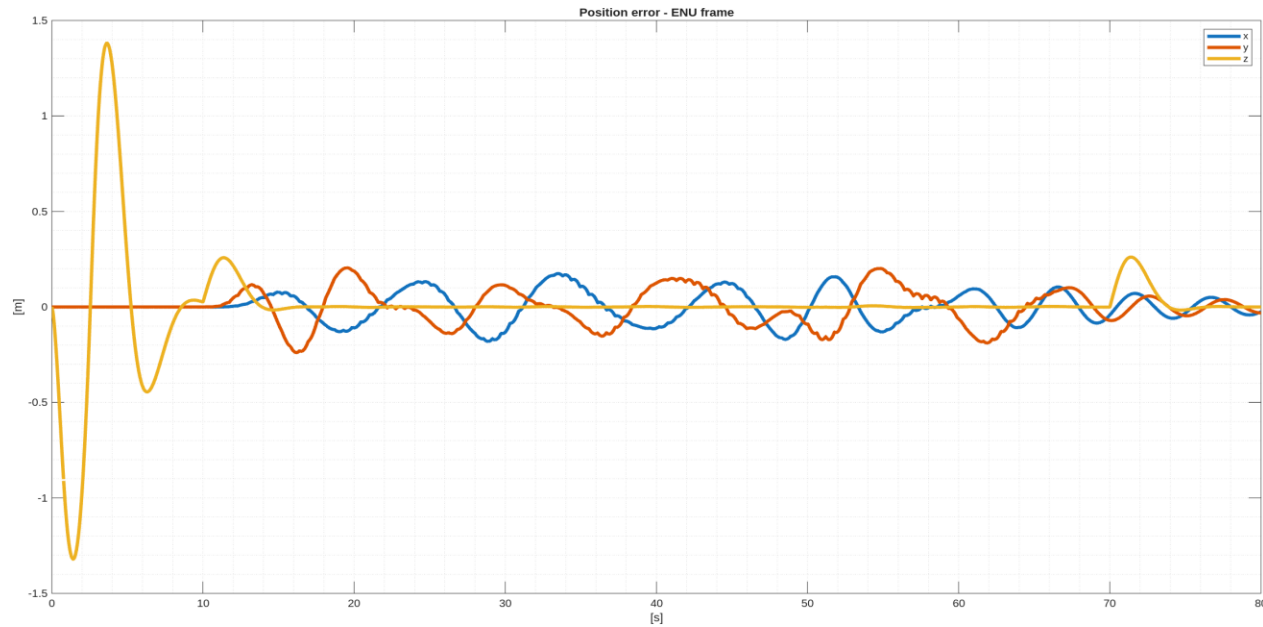


Without PB-MRAC

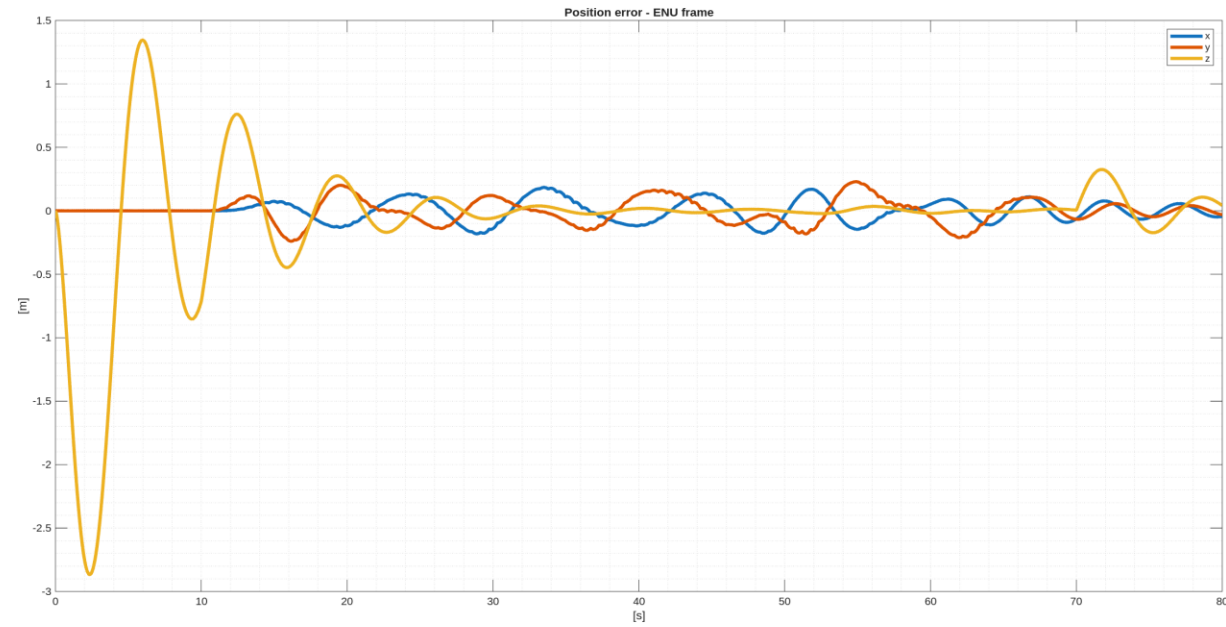


Task 4: Test in Delivery Scenario

$$\lambda_i = 0.6$$



With PB-MRAC



Without PB-MRAC

